

Final Year Project



CareFirst

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Table of Contents

1. Introduction	1
2. Vision Document	1
2.1. Problem Statement:	1
2.2. Business Opportunities	1
2.3. Objectives	1
2.4. Scope	2
2.5. Constraints	2
2.6. Stakeholder and User Description	2
2.6.1. Market Demographics	2
2.6.2. User Environment	3
2.6.3. Stakeholder Profiles	3
2.6.4. Stakeholder Summary	5
3. System Requirements Specification	5
3.1. System Features	5
3.2. Functional Requirements	6
3.2.1 Doctor Prescription	6
3.2.2 Test Search	6
3.2.3 Price Comparison	6
3.2.4 True Cost Analysis	7
3.2.5 Installment Plan	7
3.2.6 Report Upload	7
3.2.7 Translation using TTS	7
3.2.8 Financial Assistance/Full Funding	7
3.2.9 Notification Alerts	7
3.2.10 Reports & Analytics	8
3.2.11 Digital Wallet Ledger	8
3.3. Non-Functional Requirements	8
3.3.1 User-Friendly Interface	8
3.3.2 Data Security	8
3.3.3 Performance	8
3.3.4 Reliability	8
3.3.5 Compatibility	8

CareFirst

3.3.6 Maintainability	9
4. Use Case Diagram	11
5. High Level Use Case.....	12
5.1. User Login/Register	12
5.2. Search Doctor	12
5.3. Book Doctor Appointmnet	13
5.4. Search Prescribe Test.....	13
5.5. Book Lab Test	13
5.6. Manage Availibilty/Fees.....	14
5.7. Lab Listing & Comparison	14
5.8. Notification System	15
5.9. Manage Test & Pricing	15
5.10. Installment Plan	15
5.11. Manage Defaulter Case	16
5.12. Manage User-----	16
5.13. Monitor Digital Wallet-----	16
5.14. Prescribed Test-----	17
5.15. View Report-----	17
5.16. Make Payment-----	17
5.17. Upload Patient Receipt-----	18
5.18. True Cost Analysis-----	18
5.19. TTS Engine-----	18
5.20. Accept/Reject Registration-----	18
5.21. Donate-----	18
6. Expanded Use Case:.....	19
6.1. User Login/Register	19
6.2. Search Doctor	20
6.3. Book Doctor Appointment	20
6.4. Search Test	21
6.5. Book Lab Test.....	21
6.6. Lab Listing & Comparison	22
6.7. Notication System	22
6.8. Manage Test & Pricing	23

CareFirst

6.9. Installment Plan	23
6.10. Manage Defaulter Case	24
6.11. Manage User	24
6.12. Monitor Digital Wallet-----	25
6.13. Prescribed Test-----	25
6.14. View Report-----	26
6.15. Make Payment-----	26
6.16. Manage Availably & Fees-----	27
6.17. Upload Patient Receipt-----	27
6.18. True Cost Analysis-----	28
6.19. TTS Audio Conversion-----	28
6.20. Accept/Reject Registration-----	29
6.21. Donate-----	30
7. Activity Diagram.....	30
7.1 User Register/Login	31
7.2 Book Doctor Appointment	32
7.3 Bool Lab Test	33
7.4 Installment Plan.....	34
7.5 Make Payment.....	35
7.6 Manage Defaulter Case.....	36
7.7 View Report	37
7.8 Full Funded Treatment.....	38
7.9 Donate.....	39
8. Package Diagram.....	40
9. System Sequence Diagram.....	41
9.0.1. User Register/Login.....	41
9.0.2. Book Doctor Appointmnet.....	42
9.0.3. Book Lab Test	43
9.0.4. True Cost Analysis-----	44
9.0.5. Installment Plan	45
9.0.6. Manage Paymnet	46
9.0.7. Manage Deafulter Case	47
9.0.8. View Report	48

CareFirst

9.0.9. Full Funded Treatment	49
9.0.10. Donate	50
10. Class Diagram	51
11. ERD Diagram.....	52
12. Domain Diagram-----	53
References.....	55

List of Tables

Table 1:problem statement.....	1
Table 2: Patient.....	3
Table 3: Doctor	3
Table 4: Lab	4
Table 5: Admin	4
Table 6: Lawyer	4
Table 7: Stakeholder Summary.....	5
Table 8: User Login/Register	12
Table 9:Search Doctor	12
Table 10:Book Doctor Appointment.....	12
Table 11:Search Prescribe Test	13
Table 12:Book Lab Test.....	13
Table 13:Manage Availability/Fees	13
Table 14:Lab Listing & Comparison	14
Table 15:Notification System	14
Table 16:Manage Test & Pricing	14
Table 17:Installment Plan	15
Table 18:Manage Defaulter Case	15
Table 19: Manage User-----	15
Table 20: Monitor Digital Wallet-----	16
Table 21: Prescribed Test-----	16
Table 22: View Report-----	16
Table 23: Make Payment-----	17
Table 24: Upload Patient Receipt-----	17
Table 25: True Cost Analysis-----	18
Table 26: TTS Engine-----	18
Table 27: Accept/Reject Registration-----	18
Table 28: Donate-----	19
Table 29: User Login/Register	19
Table 30:Search Doctor	20
Table 31:Book Doctor Appointment.....	21
Table 32:Search Prescribe Test	22
Table 33:Book Lab Test.....	22
Table 34:Manage Availability/Fees	23
Table 35:Lab Listing & Comparison	23
Table 36:Notification System	24
Table 37:Manage Test & Pricing	24
Table 38:Installment Plan	24
Table 39:Manage Defaulter Case	25
Table 40: Manage User-----	25
Table 41: Monitor Digital Wallet-----	26
Table 42: Prescribed Test-----	26
Table 43: View Report-----	27
Table 44: Make Payment-----	27
Table 45: Upload Patient Receipt-----	28
Table 46: True Cost Analysis-----	28

CareFirst

Table 47: TTS Engine-----29

Table 48: Accept/Reject Registration-----30

Table 49: Donate-----30

List of Figures

Figure 1: Use Case Diagram.....	11
Figure 2: User Register/Login.....	31
Figure 3: Book Doctor Appointment	32
Figure 4: Prescribe Test	33
Figure 5: Book Lab Test.....	34
Figure 6: Installment Plan	35
Figure 7: Manage Payment.....	36
Figure 8: Manage Defaulter Case	37
Figure 9: View Report	38
Figure 10: Full Funded Treatment	39
Figure 11: Donate	40
Figure 12: Package Diagram.....	41
Figure 13: sq: User Register/Login	42
Figure 14: sq: Book Doctor Appointment	43
Figure 15: sq: Prescribed Test	44
Figure 16: sq: Book Lab Test	45
Figure 17.1: sq: True Cost Analysis.....	46
Figure 18: sq: Installment Plan.....	47
Figure 19: sq: Make Payment.....	48
Figure 20: sq: Manage Defaulter Case.....	49
Figure 21: sq: View Report.....	50
Figure 22: sq: Full Funded Treatment.....	51
Figure 23: sq: Donate.....	52
Figure 24: sq: Manage Test	53
Figure 25: Class Diagram.....	54
Figure 26: ERD Diagram	55
Figure 27: Domain Model Diagram	56
Figure 28: DFD Level 0	57
Figure 29: DFD Level 1	58
Figure 30: DFD Level 2.1	59
Figure 31: DFD Level 2.2	60
Figure 32: DFD Level 2.3	61
Figure 33: DFD Level 2.4	62
Figure 34: DFD Level 2.5	63
Figure 35: Sequence Diagram	64

1. Introduction

CareFirst is a web-based healthcare platform that helps patients easily find doctors, book appointments, and search for medical tests. After a doctor consultation, patients can search for prescribed tests and view all labs offering those tests along with their prices and ratings. The system uses a True Cost Analysis algorithm that considers both test price and travel cost to suggest the most affordable lab. CareFirst also provides installment payment plans for patients who cannot afford expensive tests and tracks payments through a digital ledger. Additionally, the system includes features like Text-to-Speech for reports and legal agreements, charity support through labs, and a full-funded treatment request option for eligible patients.

2. Vision Document

2.1. Problem Statement:

Table 1:problem statement

Category	Description
THE PROBLEM	Many patients struggle to find affordable labs for medical tests after a doctor consultation. Test prices vary between labs, and expensive tests create financial difficulties for patients who cannot pay the full amount.
AFFECTS	Patients may delay or avoid important medical tests due to high costs. They may also choose expensive labs because they cannot easily compare prices and travel expenses.
IMPACT	Delayed medical tests can negatively affect patient health and treatment outcomes. High test costs also create financial stress for patients.
SOLUTION	CareFirst lets patients search tests, compare lab prices with travel cost, pay in installments, translate reports via TTS, and get financial help.

2.2. Business Opportunities

CareFirst creates new opportunities in the healthcare technology sector by connecting patients, doctors, and labs on one platform. It helps labs reach more patients by displaying their services, prices, and ratings online. The system also encourages more patients to conduct medical tests because they can find affordable options easily. Installment plans allow patients to afford expensive tests, which increases service usage for labs. The platform can also grow in the future by partnering with hospitals, diagnostic centers, and healthcare organizations. Overall, CareFirst improves healthcare accessibility.

2.3. Objectives

Following are the objectives of CareFirst.

- **Doctor Prescription:** Doctors can prescribe tests to patients.

- **Test Search:** Allow patients to easily find prescribed medical tests.
- **Price Comparison:** Help patients compare lab prices and find the cheapest option.
- **Travel Cost Analysis:** Optimize Cost based on distance.
- **Installment Payments:** Enable patients to pay for tests in installments.
- **Report Translation:** Provide report translation using TTS for easy understanding.
- **Financial Assistance:** Support patients with funding or aid options.

2.4. Scope

CareFirst allows doctors to prescribe tests and helps patients search for them easily. Patients can compare lab prices, consider travel costs, and choose the most affordable option. The platform supports installment payments and translates reports using TTS for better understanding. It also provides financial assistance options for patients in need. Overall, CareFirst makes healthcare more accessible, cost-effective, and convenient for both patients and doctors.

2.5. Constraints

1. Users need internet access to use the platform smoothly, and service quality may vary by connection speed.
2. Lab options depend on regional partnerships, so availability may differ across locations.
3. Payment plans and financial assistance follow specific eligibility rules set by the platform.
4. Report translations rely on supported languages, and TTS accuracy may vary slightly.

2.6. Stakeholder and User Description

2.6.1. Market Demographics

The market for CareFirst mainly includes patients of all ages who need medical tests and want cost-effective options. Urban and semi-urban populations with access to smartphones or computers are the primary users. Doctors who prescribe tests are also part of the demographic, as they guide patients in using the platform. Partnered labs that offer affordable testing are included in the market. People seeking installment payment options or financial assistance form a key segment. Additionally, non-native language speakers benefit from report translation features. Overall, the platform targets tech-savvy patients, healthcare providers, and labs looking for convenient, efficient, and accessible medical services.

2.6.2. User Environment

CareFirst operates in a digital environment accessible via smartphones, tablets, and computers. Users interact through a simple, user-friendly interface to search tests, compare lab prices, and make payments. Doctors, patients, and lab staff access the system according to their roles. The platform supports secure data handling and online report sharing. Users can also use TTS for report translation and apply for financial assistance. Overall, it functions in an online healthcare ecosystem connecting patients, doctors, and labs efficiently.

2.6.3. Stakeholder Profiles

2.6.3.1. Patient

Table 2: Patient

Attribute	Description.
Representatives	Individuals seeking medical tests
Description	Uses platform to search tests, compare prices, and make payments
Type	External.
Responsibility	The patient must provide accurate personal and medical information, select the required tests carefully, and follow platform instructions for payments and report access.
Success Criteria	Can easily find and access affordable tests. Receives clear and accurate test reports.
Involvement	High (Patient has most involvement).
Comments/Issues	Needs support for payments and translations.

2.6.3.2. Doctor

Table 3: Doctor

Attribute	Description.
Representatives	Licensed medical professionals
Description	Manage its appointment with the patient and Prescribes tests and guides patients.
Type	External.
Responsibility	The doctor is responsible for prescribing the correct medical tests advising patients on test priorities and choices, and ensuring that patients follow proper procedures.
Success Criteria	Prescriptions are correctly followed by patients. Patients' complete tests accurately and on time.
Involvement	Medium.
Comments/Issues	Ensuring patient compliance involves clear communication of treatment plans and regular follow-up.

2.6.3.3. Lab

Table 4: Lab

Attribute	Description
Representatives	Lab technicians and managers
Description	Conducts tests and uploads reports
Type	Internal
Responsibility	Lab staff must perform accurate lab tests, upload reports promptly to the system, and maintain lab equipment and data records properly.
Success Criteria	Test results are accurate and timely. Reports are delivered reliably to patients and doctors.
Involvement	High (daily task execution and reporting).
Comments/Issues	Manage lab availability and data entry.

2.6.3.4. Admin

Table 5: Admin

Attribute	Description
Representatives	Platform administrators.
Description	Maintains platform, manages users and labs.
Type	Internal
Responsibility	The admin is responsible for managing user accounts and roles, ensuring system security and smooth operation, and resolving technical issues or disputes.
Success Criteria	Platform runs without downtime. Users experience a secure and stable system.
Involvement	High (Manage Platform).
Comments/Issues	The admin makes sure patients follow instructions and attend appointments. They also solve disputes and fix data problems quickly.

2.6.3.5. Lawyer

Table 6: Admin

Attribute	Description
Representatives	Legal professionals.
Description	Ensures compliance with laws and handles disputes.
Type	External.
Responsibility	The lawyer reviews contracts and agreements to ensure they follow healthcare and data laws. They also handle disputes, send legal notices to defaulting patients, and manage all legal contracts.
Success Criteria	Platform complies with all regulations. Legal disputes are resolved efficiently and fairly.
Involvement	Low (active during Legal Terminology).
Comments/Issues	Handles data privacy and patient/legal disputes manage all the confidential paper work.

2.6.4. Stakeholder Summary

Table 7: Stakeholder Summary

Name	Description	Responsibility
Patient	Individual seeking medical tests and using the platform for search and payments.	Provide accurate personal and medical information. Select required tests carefully and on time. Follow platform instructions for payments and report access.
Doctor	Licensed medical professional who prescribes tests and guides patients and also track the patient previous medical history.	The role includes prescribing the right medical tests and setting priorities for patient care. It also involves advising patients on proper procedures and guiding them to follow instructions. Finally, it ensures patients comply with care plans for effective treatment.
Lab	Technician and managers conducting test and uploading report.	The role involves performing accurate lab tests by following standard procedures carefully. Reports are uploaded promptly into the system to ensure timely access. It also includes maintaining lab equipment properly and keeping data records organized and correct.
Admin	Platform administrator managing users, labs, and system operations.	Manage user accounts and roles effectively. Ensure system security and smooth operation. Resolve technical issues and user disputes efficiently.

Lawyer	Legal professional ensuring compliance and handling disputes	The role requires reviewing legal agreements and contracts carefully to ensure they comply with healthcare and data laws. It also involves handling legal disputes or patient complaints in a professional manner, safeguarding both organizational and patient interests.
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3. System Requirements Specification

3.1. System Features

Project offers a range of critical features, including:

- **Doctor Prescription:** Doctors can prescribe tests directly to patients. This ensures accurate guidance and proper test selection.
- **Test Search:** Patients can easily search for prescribed or general medical tests. Search is organized by keywords or test categories.
- **Price Comparison:** Compares costs of tests across multiple labs. Highlights the most affordable and suitable options for patients.
- **True Cost Analysis:** Suggests labs based on distance and travel costs. Helps patients choose labs that are convenient and economical.
- **Installment Payment:** Enables patients to pay for expensive tests in multiple installments. Ensures affordable and flexible payment options.
- **Report Upload & Access:** Labs can upload test reports to the system. Patients can securely view or download reports anytime.
- **Translation using TTS:** Converts reports and all the legal documentation into audio or local language formats. Improves understanding for all patients, including non-native speakers.

- **Financial Assistance Management:** Supports patients applying for aid and tracks approval status. Helps patients afford tests they cannot pay fully.
- **Notifications & Alerts:** Sends updates about test status, reports, and financial aid. Keeps patients informed and engaged throughout the process.
- **Reports & Analytics:** Admins can track lab performance, user activity, and system usage. Helps improve operations and decision-making.
- **Digital Wallet / Ledger:** Tracks all patient payments and installments securely. Provides a clear record of transactions for transparency.

3.2. Functional Requirements

The system must fulfil specific functional requirements, including:

3.2.1 Doctor Prescription

- Doctors can prescribe medical tests digitally to patients, ensuring they follow accurate guidance for their treatment.
- Prescriptions are recorded in the system and can be updated as patient conditions change over time.
- Enables doctors to monitor patient compliance and track their test history effectively

3.2.2 Test Search

- Patients can search for medical tests by name, category, or relevant keywords using an intuitive interface.
- Search results display available labs along with prices and locations for easy comparison.
- Filters allow patients to sort by lab ratings, distance, and cost to choose the most suitable option.

3.2.3 Price Comparison

- The system compares the cost of the same test across multiple partnered labs automatically.
- Highlights the cheapest and most suitable option to help patients save money.
- Prices are updated dynamically whenever labs modify their rates, ensuring accurate recommendations.

3.2.4 True Cost Analysis

- Calculates estimated travel distance and cost from the patient's location to each lab.
- Suggests labs that minimize both travel time and overall expenses for convenience.
- Helps patients choose a balance between affordability and accessibility for their tests.

3.2.5 Installment Payments

- Enables patients to pay for expensive tests in multiple installments securely and efficiently.
- Tracks payment status, updates balance in real-time, and maintains transaction history.
- Sends automated reminders for upcoming installments to ensure timely payments.

3.2.6 Report Upload

- Labs can upload test reports to the system securely, allowing patients to access them online.
- Patients can view its report conveniently through the platform.

3.2.7 Translation using TTS

- . Converts test reports into local languages for patients with language or accessibility needs.
- Patients can see reports or read them in their preferred language for better understanding.
- Patient can easily understand reports or all the legal agreements in his native language.

3.2.8 Financial Assistance / Full Funding

- Patients can submit support documents which can help him to apply for full or partial financial aid if he is not able to afford his expensive.
- The admin will evaluate eligibility and approves funding based on criteria
- Tracks the approval status and notifies patients once aid is granted

3.2.9 Notifications & Alerts

- Sends timely updates about test status, report availability, payment reminders, and financial aid notifications.
- Patient, doctor and Lawyer receive alerts via email, SMS, or in-app notifications.

3.2.10 Reports & Analytics

- Admin can view all patients who requested financial assistance or full funding.
- Admin can see detailed information, including submitted documents, eligibility and then give approval status.

3.2.11 Digital Wallet / Ledger

- Maintains a secure ledger to record all patient payments and installment transactions accurately.
- When an installment is paid, the ledger updates and schedules the reminder for next installment.

3.3. Non-Functional Requirements

3.3.1 User-Friendly Interface

- The system should be simple and easy for patients, doctors, and lab staff to use, with clear menus and options, so users can complete tasks quickly without confusion.

3.3.2 Data Security

- All patient data, lab, and payment data should be safe and protected from unauthorized access, ensuring privacy and compliance with laws.

3.3.3 Performance

- The system should respond quickly to actions like searching tests, uploading reports, or comparing prices, and work smoothly even when many users use it at the same time.

3.3.4 Reliability

- The system should be available almost all the time without crashes, and all reports, test results, and payment records must be stored and retrieved correctly.

3.3.5 Compatibility

- The system should work well on computers, tablets, and smartphones, and be supported by all major browsers so everyone can access it easily.

3.3.6 Maintainability

- The system should be easy to fix, update, and add new features, allowing developers to manage it without causing problems for users.

4. Use Case Diagram

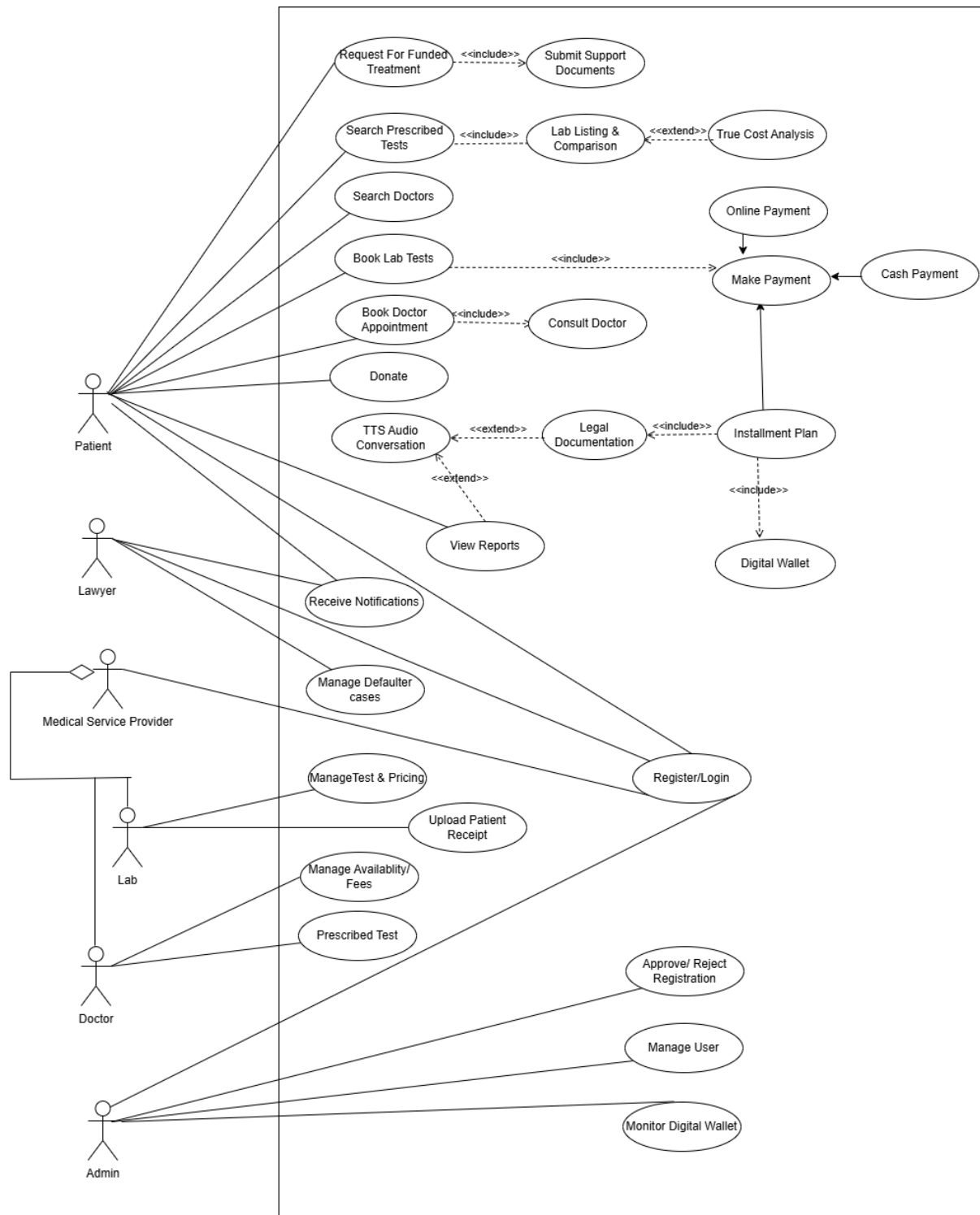


Figure 1: Use Case Diagram

5. High Level Use Case

5.1. User Registration/Login

Table 8: User Registration/Login

Attribute	Detail
Use Case ID	UC-01
Use Case Name	Registration/Login
Actor(s)	Patient, Doctor, Lab, Lawyer and Admin
Type	Primary
Description	Patient, Doctor, Lab, Admin and Lawyer registers in the system by providing personal and contact information.

5.2. Search Doctor

Table 9: Search Doctor

Attribute	Detail
Use Case ID	UC-02
Use Case Name	Search Doctors
Actor(s)	Patient
Type	Primary
Description	Patient can easily search for doctors based on specialty or field of the doctors.

5.3. Book Doctor Appointment

Table 10: Book Doctor Appointment

Attribute	Detail
Use Case ID	UC-03
Use Case Name	Book Doctor Appointment
Actor(s)	Patient
Type	Primary

Description	Patient can view the profile of the doctor and schedules an appointment in the available slot of a doctor.
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5.4. Search Prescribed Test

Table 11: Search Prescribed Test

Attribute	Detail
Use Case ID	UC-04
Use Case Name	Search Prescribed Tests
Actor(s)	Patient
Type	Secondary
Description	Patient can easily search for lab tests which is prescribed to patient by the doctors.

5.5. Book Lab Test

Table 12: Book Lab Test

Attribute	Detail
Use Case ID	UC-05
Use Case Name	Book Lab Test
Actor(s)	Patient
Type	Primary
Description	Patient can easily book lab tests at a preferred lab among from the listing of the labs.

5.6. Manage Availability / Fees

Table 13: Manage Availability / Fees

Attribute	Detail
Use Case ID	UC-06
Use Case Name	Manage Availability / Fees
Actor(s)	Doctor, Lab.
Type	Primary
Description	Doctors manage appointment slots and fees.

5.7. Lab Listing & Comparison

Table 14: Lab Listing & Comparison

Attribute	Detail
Use Case ID	UC-07
Use Case Name	Lab Listing & Comparison
Actor(s)	Patient.
Type	Secondary.
Description	System lists labs offering a test and allows price comparison for transparency.

5.8. Notification System

Table 15: Notification system

Attribute	Detail
Use Case ID	UC-08
Use Case Name	Receive Notifications
Actor(s)	Doctor, Lawyer, Patient, Lab, Admin.
Type	Primary
Description	Users receive system notifications and updates.

5.9. Manage Test & Pricing

Table 16: Manage Test & Pricing

Attribute	Detail
Use Case ID	UC-09
Use Case Name	Manage Test & Pricing
Actor(s)	Lab.
Type	Secondary
Description	Lab staff manage available tests and pricing details.

5.10. Installment Plan

Table 17: Installment Plan

Attribute	Detail
Use Case ID	UC-10
Use Case Name	Installment Plan
Actor(s)	Patient
Type	Secondary
Description	Allows patients to pay treatment cost in installments.

5.11. Manage Defaulter Case

Table 18: Manage Defaulter Case

Attribute	Detail
Use Case ID	UC-11
Use Case Name	Manage Defaulter Case
Actor(s)	Lawyer
Type	Secondary.
Description	Lawyer handle and manages the defaulter patient case if he fails to pay the installments.

5.12. Manage User

Table 19: Manage User

Attribute	Detail
Use Case ID	UC-12
Use Case Name	Manage User
Actor(s)	Admin
Type	Primary.
Description	Admin manages all types of user accounts and their roles and what access should they have.

5.13. Monitor Digital Wallet

Table 20: Monitor Digital Wallet

Attribute	Detail
Use Case ID	UC-13
Use Case Name	Monitor Digital Wallet
Actor(s)	Admin, Patient.
Type	Secondary.
Description	Users track all of the transactions and the remaining installments and pending dues in the system wallet.

5.14. Prescribed Test

Table 21: Prescribed Test

Attribute	Detail
Use Case ID	UC-14
Use Case Name	Prescribed Test
Actor(s)	Doctor
Type	Secondary.
Description	Doctor examines the patient completely and prescribes the best tests for the patient after the consultations.

5.15. View Report

Table 22: View Report

Attribute	Detail
Use Case ID	UC-15
Use Case Name	View Report
Actor(s)	Patient, Doctor.
Type	Secondary.
Description	Patient and Doctor can view the report of the patient after the test reports upload by the lab.

5.16. Make Payment

Table 23: Make Payment

Attribute	Detail
Use Case ID	UC-16
Use Case Name	Make Payment
Actor(s)	Patient.
Type	Primary.
Description	Patient should make payment to lab or doctor after its appointment.

5.17. Upload Patient Receipt

Table 24: Upload Patient Receipt

Attribute	Detail
Use Case ID	UC-17
Use Case Name	Upload Patient Receipt.
Actor(s)	Doctor
Type	Secondary.
Description	After booking an appointment with the doctor. The doctor should upload the receipt of payment and treatment proof. .

5.18. True Cost Analysis

Table 25: True Cost Analysis

Attribute	Detail
Use Case ID	UC-18
Use Case Name	True Cost Analysis
Actor(s)	Patient
Type	Primary
Description	After checking the test patient can allow live location which calculates the distance and tells the lab which cost cheapest. .

5.19. TTS Engine

Table 26: TTS Engine

Attribute	Detail
Use Case ID	UC-19
Use Case Name	TTS Engine
Actor(s)	Patient, Doctor
Type	Secondary
Description	After seeing the report patient can request TTS engine to convert the report into native language to understand easily. .

5.20. Accept/Reject Registration

Table 27: Accept/Reject Registration

Attribute	Detail
Use Case ID	UC-20
Use Case Name	Accept/Reject Registration
Actor(s)	Admin
Type	Primary
Description	Admin approves or rejects new user registration of users. .

5.21. Donate

Table 28: Donate

Attribute	Detail
Use Case ID	UC-21
Use Case Name	Donate
Actor(s)	Users
Type	Secondary
Description	Users can donate to the labs which accepts the charity. .

6. Expanded Use Case:

6.1. User Registration/Login

Table 29: User Registration/Login

Use Case ID	UC-01
Use Case Name	User Registration/Login
Actor	Patient, Doctor, Lab Staff, Admin
Type	Primary
Description	Users create an account or log in to access the system but if don't have account need to first register.
Trigger	User select register or login option.
Pre-condition	User has internet access and system is available.
Post-condition	User account is created or user is logged into the system.

Normal Flow	1. User opens system 2. User selects register or login 3. User enters credentials 4. System verifies information 5. User successfully enters the system
Alternative Flow	1. Invalid credentials → system shows error message

6.2. Search Doctor

Table 30: Search Doctor

Use Case ID	UC-02
Use Case Name	Search Doctor
Actor	Patient
Type	Primary
Description	Patient searches doctors based on specialty or location.
Trigger	User searches for a doctor
Pre-condition	Patient login and doctor exist in the system.
Post-condition	System displays list of doctors.
Normal Flow	1. Patient enters doctor name or specialty 2. System searches database 3. System displays available doctors
Alternative Flow	1. No doctor found → system shows no results

6.3. Book Doctor Appointment

Table 31: Book Doctor Appointment

Use Case ID	UC-03
Use Case Name	Book Doctor Appointment
Actor	Patient

Type	Secondary
Description	Patient books an appointment with a doctor.
Trigger	Patient selects a doctor.
Pre-condition	Doctor is available.
Post-condition	Appointment is scheduled.
Normal Flow	1. Patient selects doctor 2. Patient selects date and time 3. System confirms booking
Alternative Flow	1. Time slot unavailable → patient selects another time

6.4. Search Prescribe Test

Table 32: Search Prescribe Test

Use Case ID	UC-04
Use Case Name	Search Prescribed Test
Actor	Patient.
Type	Secondary
Description	Patient searches tests prescribed by doctor after his consultation.
Trigger	Patient enters test name.
Pre-condition	Test exists in the system.
Post-condition	System displays available test.
Normal Flow	1. Patient enters test name 2. System searches database 3. System displays tests
Alternative Flow	1. Test not found → system shows message

6.5. Book Lab Test

Table 33: Book Lab Test

Use Case ID	UC-05
Use Case Name	Book Lab Test

Actor	Patient
Type	Primary
Description	Patient books a lab test from available labs.
Trigger	Patient selects lab test.
Pre-condition	Lab and test exist in system.
Post-condition	Lab test booking is confirmed.
Normal Flow	1. Patient selects test 2. System shows labs 3. Patient selects lab 4. Booking confirmed
Alternative Flow	1.Lab unavailable → system shows other labs

6.6. Manage Availability / Fees

Table 34: Manage Availability/Fees

Use Case ID	UC-06
Use Case Name	Manage Availability / Fees
Actor	Patient
Type	Secondary
Description	Doctor manages availability schedule and consultation fee.
Trigger	Doctor updates schedule.
Pre-condition	Doctor logged in.
Post-condition	Updated schedule stored in system.
Normal Flow	1. Doctor opens availability section 2. Doctor updates schedule and fee 3. System saves changes
Alternative Flow	1. Invalid input → system shows error.

6.7. Lab Listing & Comparison

Table 35: Lab Listing & Comparison

Use Case ID	UC-07
Use Case Name	Lab Listing & Comparison
Actor	Patient
Type	Secondary
Description	System shows labs offering a test and allows comparison.
Trigger	Patient searches test.
Pre-condition	Labs available.
Post-condition	Patient selects preferred lab.
Normal Flow	1. System lists labs 2. Patient compares price and rating 3. Patient selects lab
Alternative Flow	1. No labs available

6.8. Notification System

Table 36: Notification System

Use Case ID	UC-08
Use Case Name	Notification System
Actor	Patient, Doctor, Lab, Lawyer, Admin.
Type	Secondary
Description	System sends alerts for appointments, reports and payments.
Trigger	System event occurs.
Pre-condition	User registered.
Post-condition	Notification sent to user.

Normal Flow	1. Event occurs 2. System generates notification 3. Notification sent to user.
Alternative Flow	1. Notification failure → retry sending

6.9. Manage Test & Pricing

Table 37: Manage Test & Pricing

Use Case ID	UC-09
Use Case Name	Manage Test & Pricing
Actor	Medical Provider
Type	Secondary
Description	Lab staff manages tests and their prices.
Trigger	Lab updates test information.
Pre-condition	Lab logged in.
Post-condition	Test information updated.
Normal Flow	1. Lab opens test management 2. Lab updates test or price 3. System saves changes
Alternative Flow	1. Invalid data → system shows error

6.10. Installment Plan

Table 38: Installment Plan

Use Case ID	UC-10
Use Case Name	Installment
Actor	Patient
Type	Secondary
Description	Patient wants to pay its medical expenses in installments.
Trigger	User select installment.

Pre-condition	Patient pay down payment.
Post-condition	Installment plan and wallet activated.
Normal Flow	1. User selects installment 2. All legal work done 3. installments generated in wallet
Alternative Flow	1. Use other payments option

6.11. Manage Defaulter Case

Table 39: Manage Defaulter Case

Use Case ID	UC-11
Use Case Name	Manage Defaulter Case
Actor	Lawyer
Type	Secondary
Description	Admin Lists user who fails to pay installments and that list is sent to the lawyer for the legal work progress.
Trigger	Installment payment missed.
Pre-condition	Installments are activated in wallet.
Post-condition	Defaulter case recorded and lawyer can take legal action.
Normal Flow	1. System detects missed payment 2. Admin reviews case 3. Case forward to lawyer 4. Lawyer takes legal action
Alternative Flow	1. Payment received later → case closed

6.12. Manage User

Table 40: Manage User

Use Case ID	UC-12
Use Case Name	Manage User
Actor	Admin
Type	Primary
Description	Admin manages system users.
Trigger	Admin selects user management.
Pre-condition	Admin logged in.
Post-condition	User information and access updated.
Normal Flow	1. Admin views user list 2. Admin updates or removes user 3. System saves changes
Alternative Flow	1. User not found → system shows message

6.13. Monitor Digital Wallet

Table 40: Monitor Digital Wallet

Use Case ID	UC-13
Use Case Name	Monitor Digital Wallet
Actor	Admin
Type	Primary
Description	Admin monitors digital wallet installment plans.
Trigger	Admin opens digital wallet section.
Pre-condition	Digital Wallet exists.
Post-condition	All transaction records viewed.
Normal Flow	1 Admin opens digital wallet 2 System displays plans 3 Admin monitors activity
Alternative Flow	1. No digital wallet found

6.14. Prescribed Test

Table 41: Prescribe Test

Use Case ID	UC-14
Use Case Name	Prescribed Test
Actor	Doctor
Type	Primary
Description	Doctor prescribes medical tests to patient.
Trigger	Doctor prescribes test id necessary.
Pre-condition	Patient Consultation completed
Post-condition	Test prescription written.
Normal Flow	1.Patient selects doctor 2. Consultation completed 3. Doctor prescribes test
Alternative Flow	1. No test

6.15. View Report

Table 42: View Report

Use Case ID	UC-15
Use Case Name	View Report
Actor	Patient, Doctor
Type	Primary
Description	Patient and doctor view the test reports uploaded by the lab after the test conducted.
Trigger	Selects view report.
Pre-condition	Report uploaded.
Post-condition	Report displayed.
Normal Flow	1. Patient opens report section 2. System retrieves report 3. Report displayed
Alternative Flow	1. Pending → Report not available

6.16. Make Payment

Table 43: Make Payment

Use Case ID	UC-16
Use Case Name	Make Payment
Actor	Patient
Type	Primary
Description	Patient pays consultation or test fee online, cash or installment plan.
Trigger	Patient selects payment option.
Pre-condition	Payment required.
Post-condition	Payment receipt uploaded and recorded.
Normal Flow	1. Patient selects payment method 2. Patient processes payment 3. Payment confirmed 4. System notified
Alternative Flow	1. Not received → try again

6.17. Upload Receipt

Table 44: Upload Receipt

Use Case ID	UC-17
Use Case Name	Upload Receipt
Actor	Lab
Type	Primary
Description	Lab uploads receipt after patient 'tests.
Trigger	Lab selects upload receipt option.
Pre-condition	Test is conducted.
Post-condition	Patient receipt is ready to upload.
Normal Flow	1. Lab uploads receipt file 2. System notifies patient 3. Patient view receipt
Alternative Flow	1. → File upload error

6.18. True Cost Analysis

Table 45: True Cost Analysis

Use Case ID	UC-18
Use Case Name	True Cost Analysis
Actor	Patient
Type	Primary

Description	System calculates total cost of test including additional factors like traveling by taking user live location.
Trigger	Patient selects lab test.
Pre-condition	Test selected.
Post-condition	Final lab with cheapest cost displayed.
Normal Flow	1. User select lab. 2. System ask live location. 3. Apply true cost analysis. 4. Display the lab which cost cheapest.
Alternative Flow	1. Select lab without true cost analysis

6.19. TTS Audio Conversion

Table 46: TTS Audio Conversion

Use Case ID	UC-19
Use Case Name	TTS Audio Conversation
Actor	Patient, Doctor, Lawyer.
Type	Secondary.
Description	System converts text information into audio in your native language to easily understand.
Trigger	User selects audio option.
Pre-condition	Report and legal documentation content exists.
Post-condition	Audio generated.
Normal Flow	1. User selects report or legal documents 2. System converts to speech in native language 3. Audio plays
Alternative Flow	1. Audio generation error.

6.20. Approve/Reject Registration

Table 47: Approve and Reject Registration

Use Case ID	UC-20
Use Case Name	Approve and Reject Registration
Actor	Admin
Type	Primary
Description	Admin approves or rejects user registration requests from users and labs.
Trigger	New registration request submitted.

Pre-condition	User registration pending.
Post-condition	Registration approved/rejected and user notified.
Normal Flow	1. Admin views registration request 2. Approves or rejects 3. System updates status
Alternative Flow	1. User info incomplete → request rejected

6.21. Donate

Table 48: Donate

Use Case ID	UC-21
Use Case Name	Donate
Actor	Patient, Doctor, Lawyer.
Type	Secondary.
Description	Users can donate to the labs who accept charity and use to favor the patient who can't afford fee.
Trigger	User selects donation option.
Pre-condition	User logged in.
Post-condition	User donate charity.
Normal Flow	1. User want to donate 2. Labs list who accept charity 3. User donates 4.system notified
Alternative Flow	1. Payment failure → retry payment

7. Activity Diagram

7.1 Register/Login

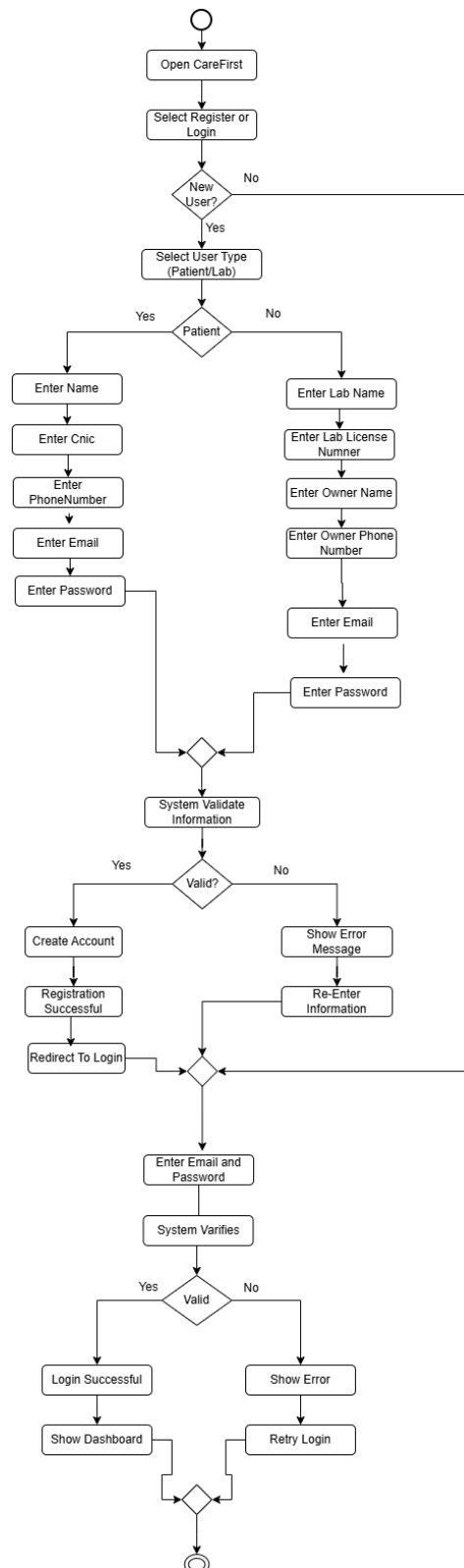


Figure 2: Register/Login

7.2. Doctor Appointment

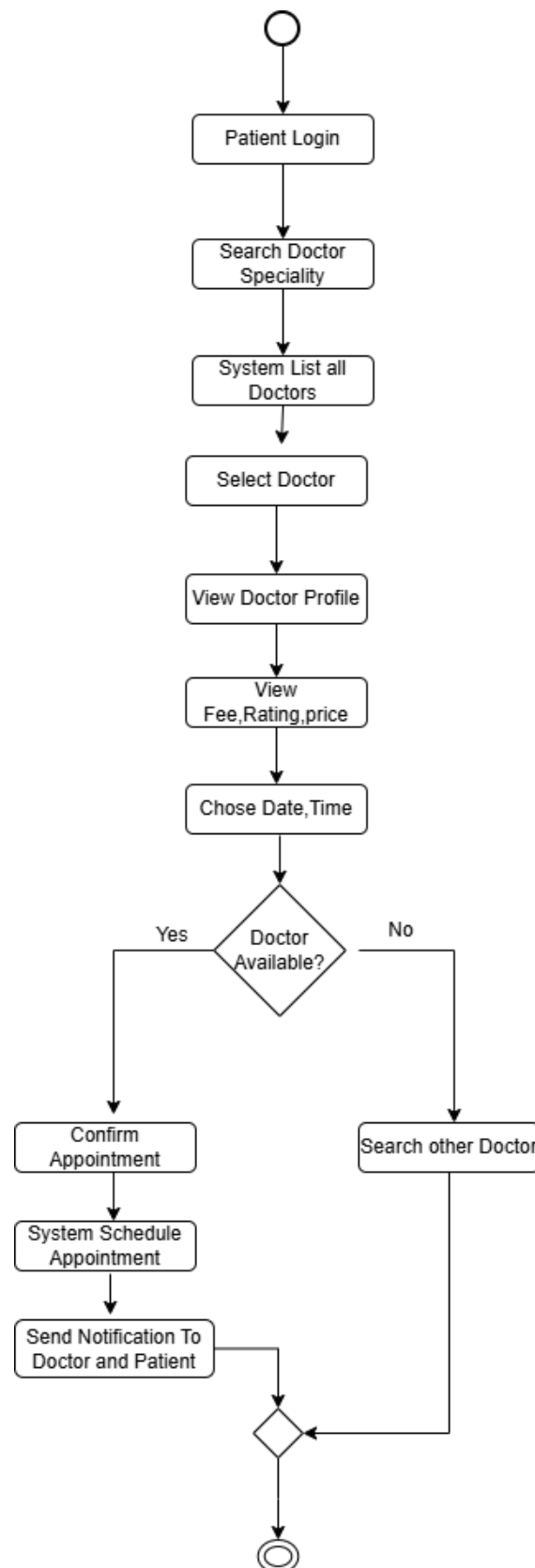


Figure 3: Doctor Appointment

7.3. Prescribed Test

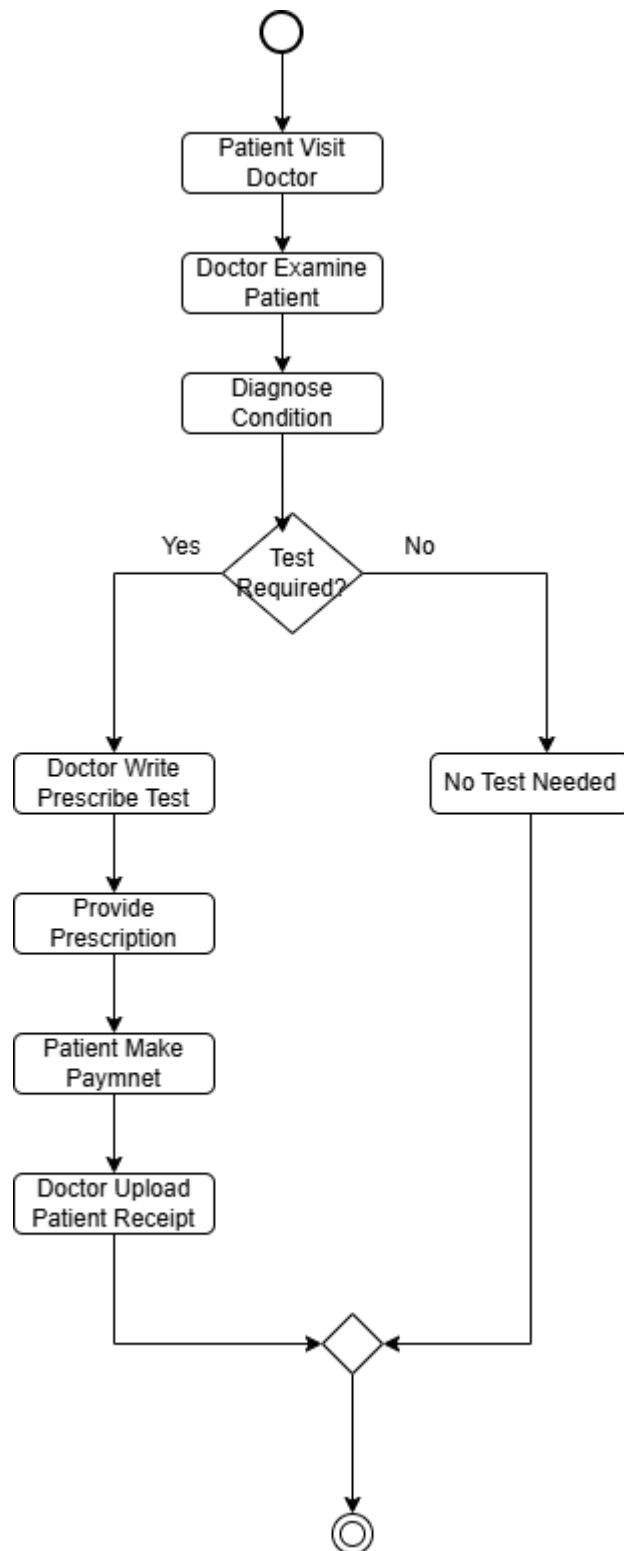


Figure 4: Prescribed Test

7.4. Book Lab Test

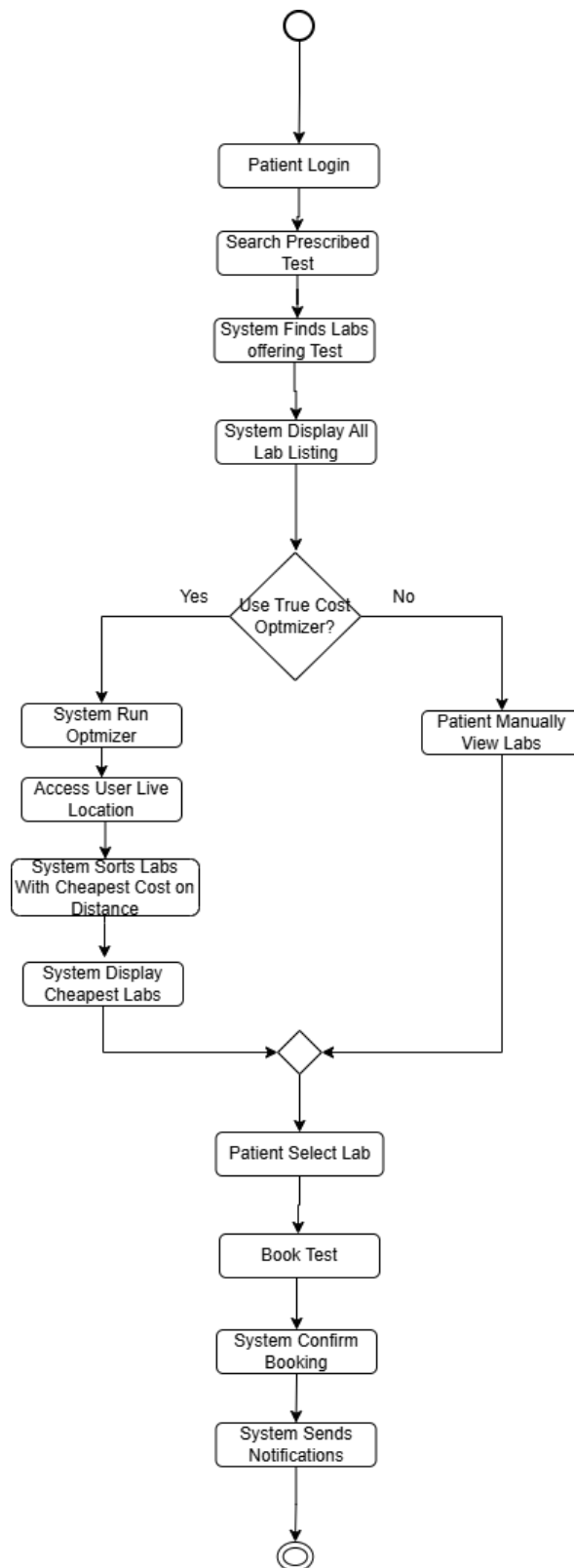


Figure 5: Book Lab Test

7.5. Installment Plan

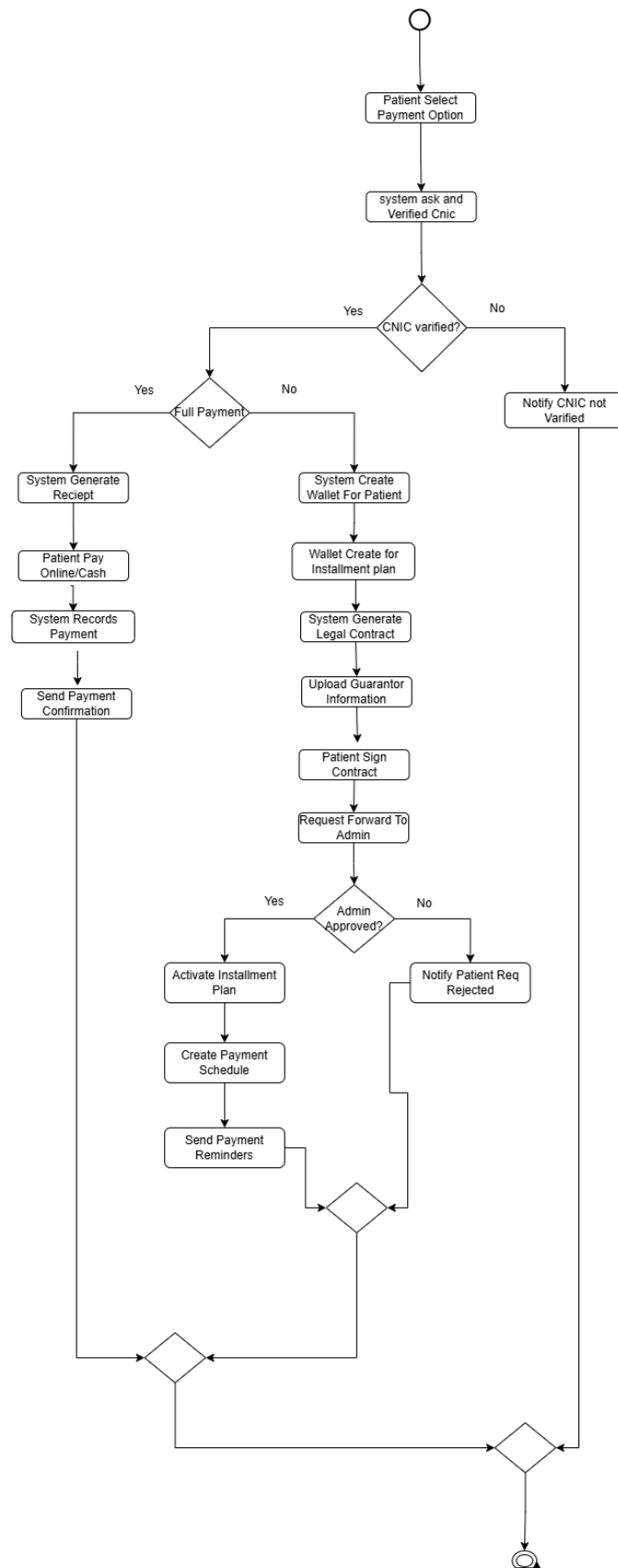


Figure 6: Installment Plan

7.6. Make Payment

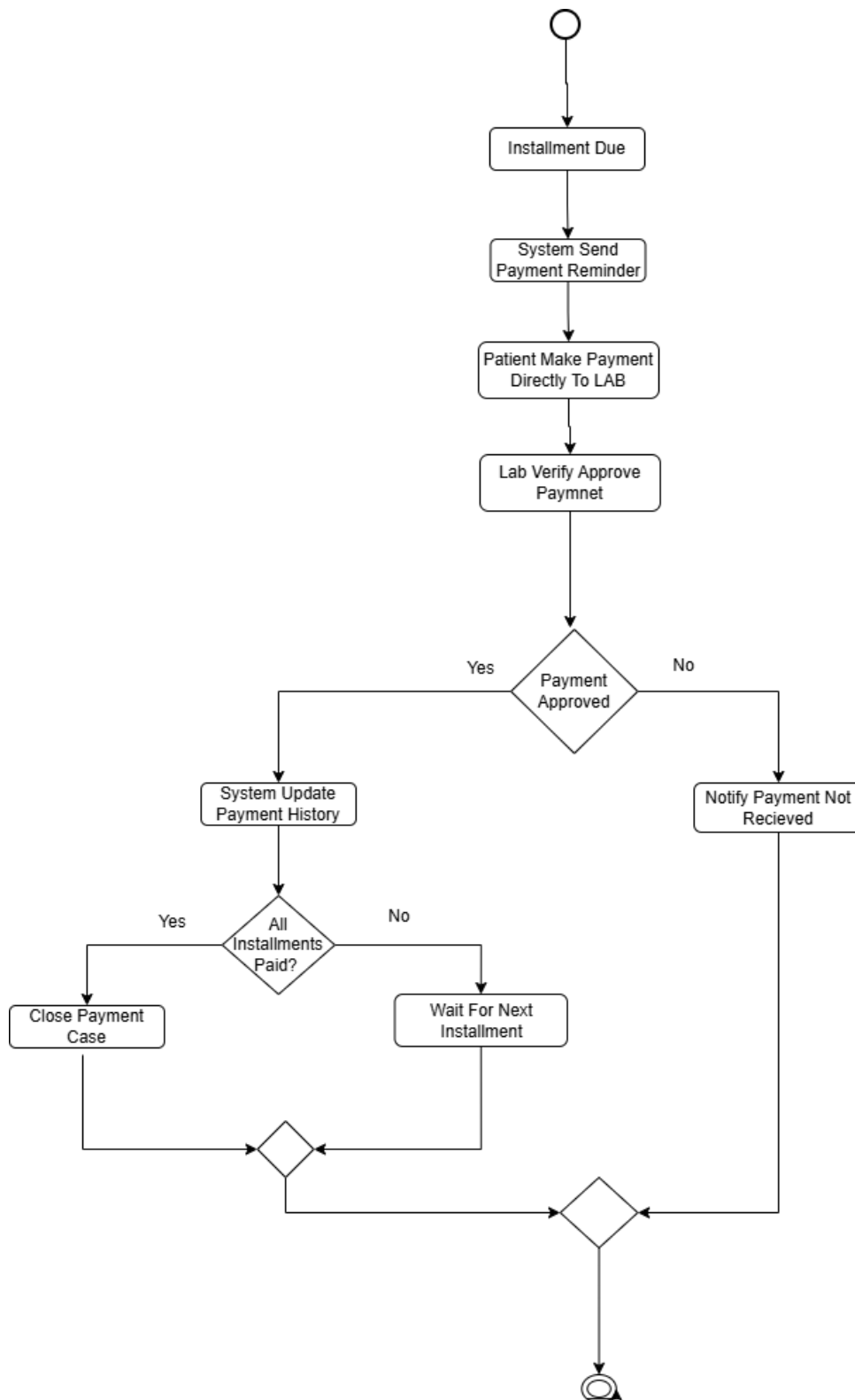


Figure 7: Make Payment

7.7. Manage Defaulter Case

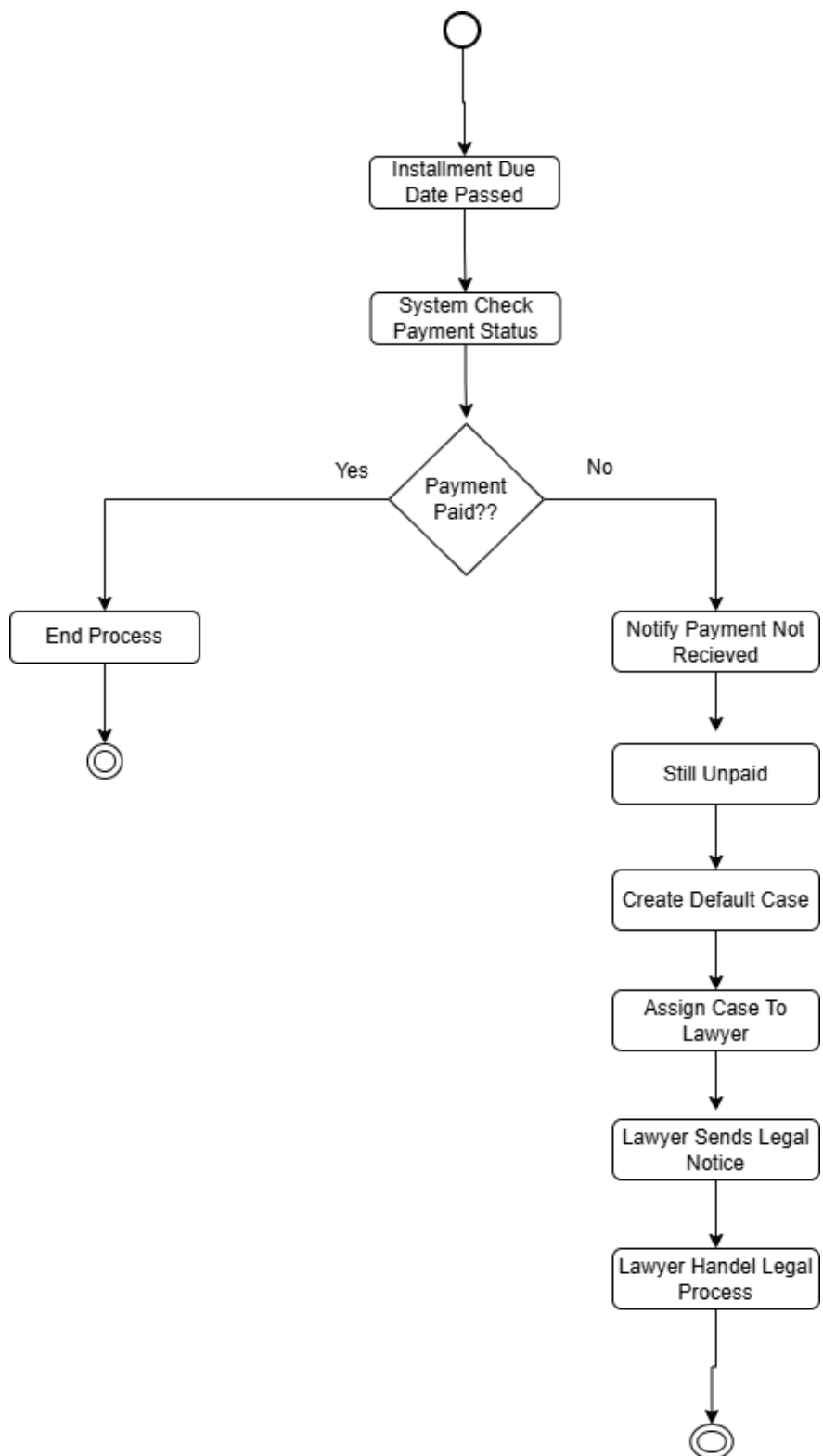


Figure 8: Manage Defaulter Case

7.8. View Report

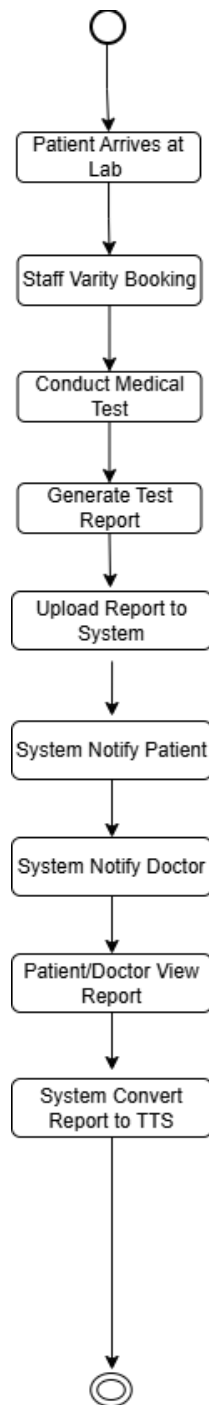


Figure 9: View Report

7.9. Full Funded Treatment

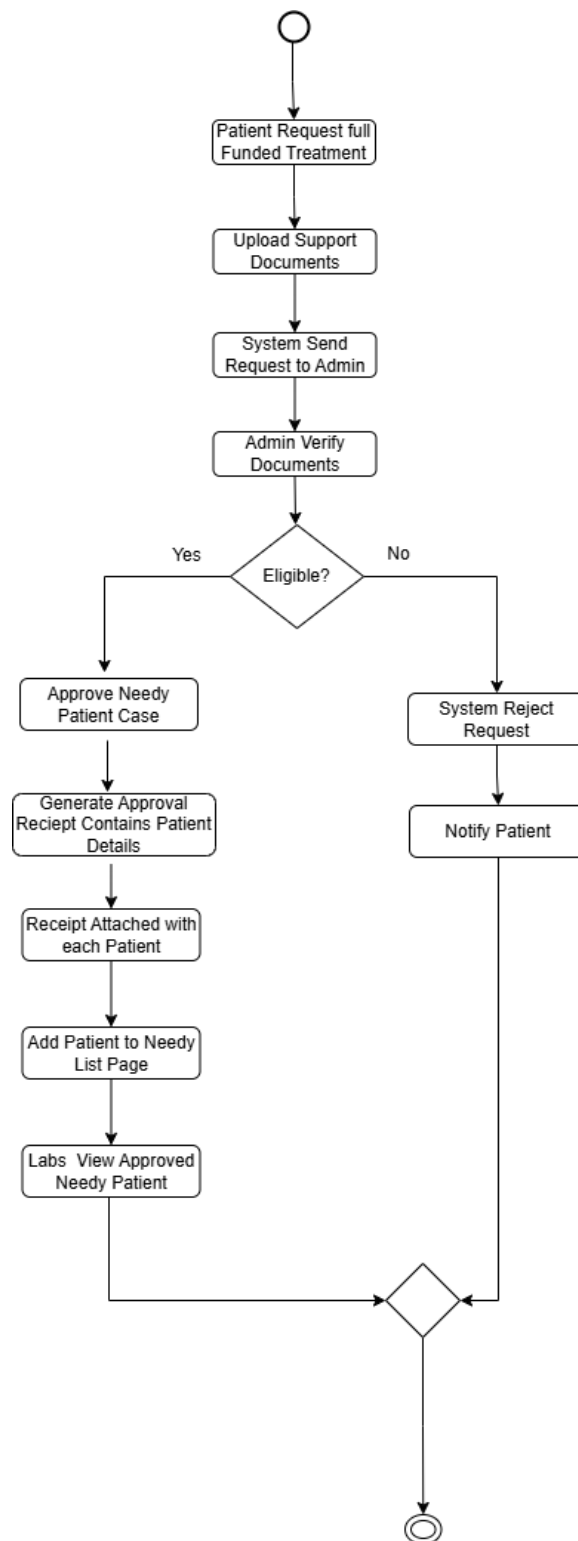


Figure 10: Full Funded Treatment

7.10. Donate

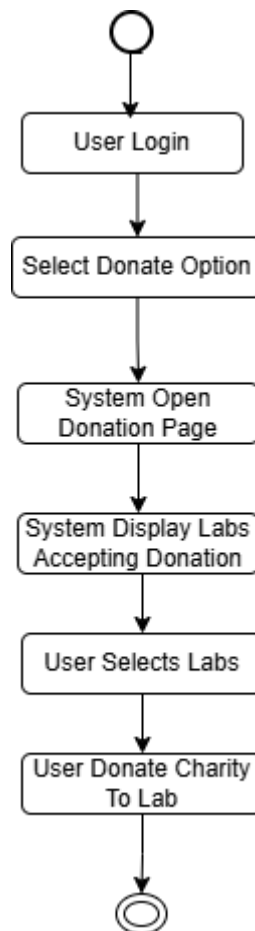


Figure 11: Donate

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8. Package Diagram

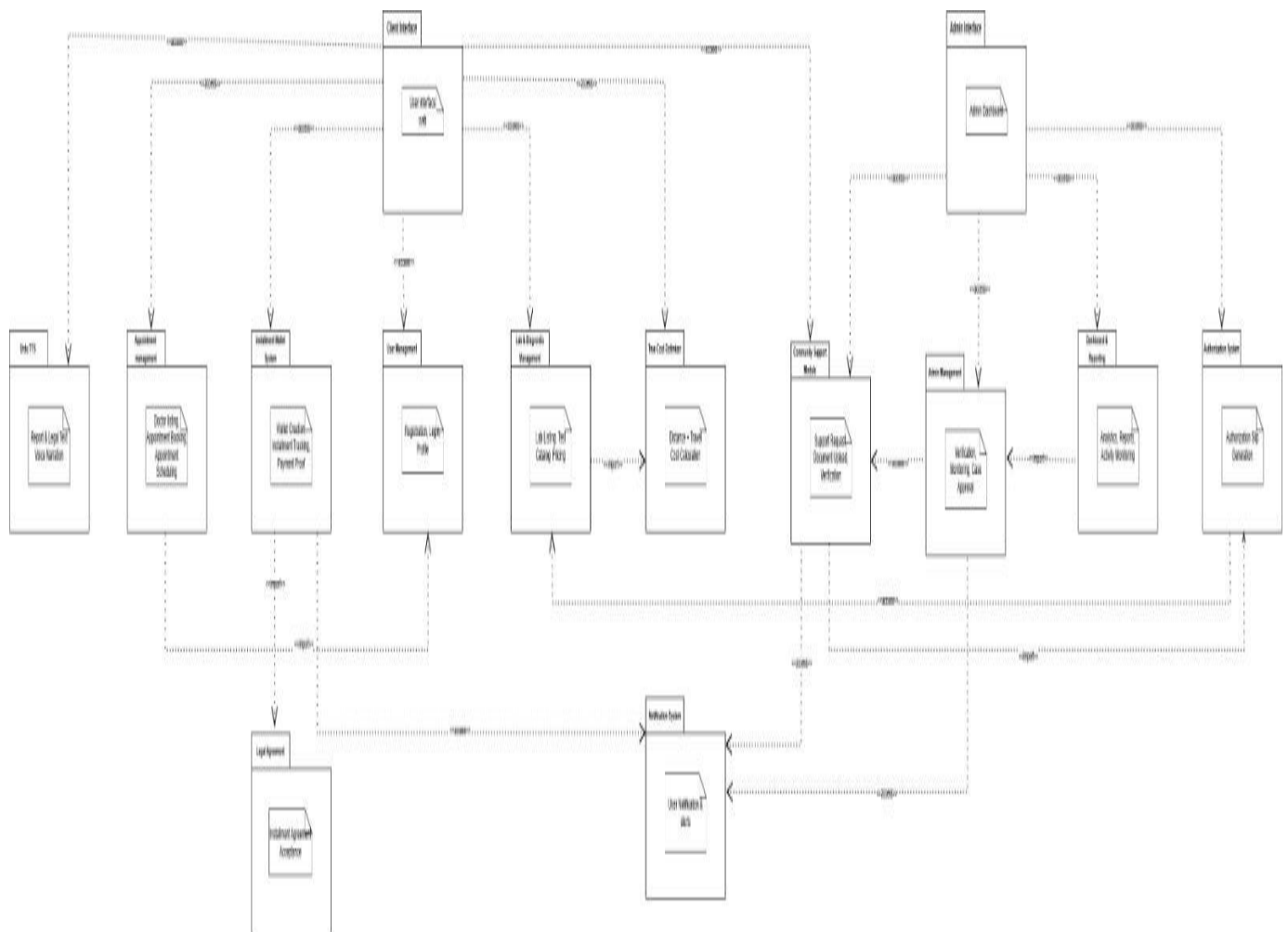


Figure 12: Package Diagram

9. System Sequence Diagram

9.0. Register/Login

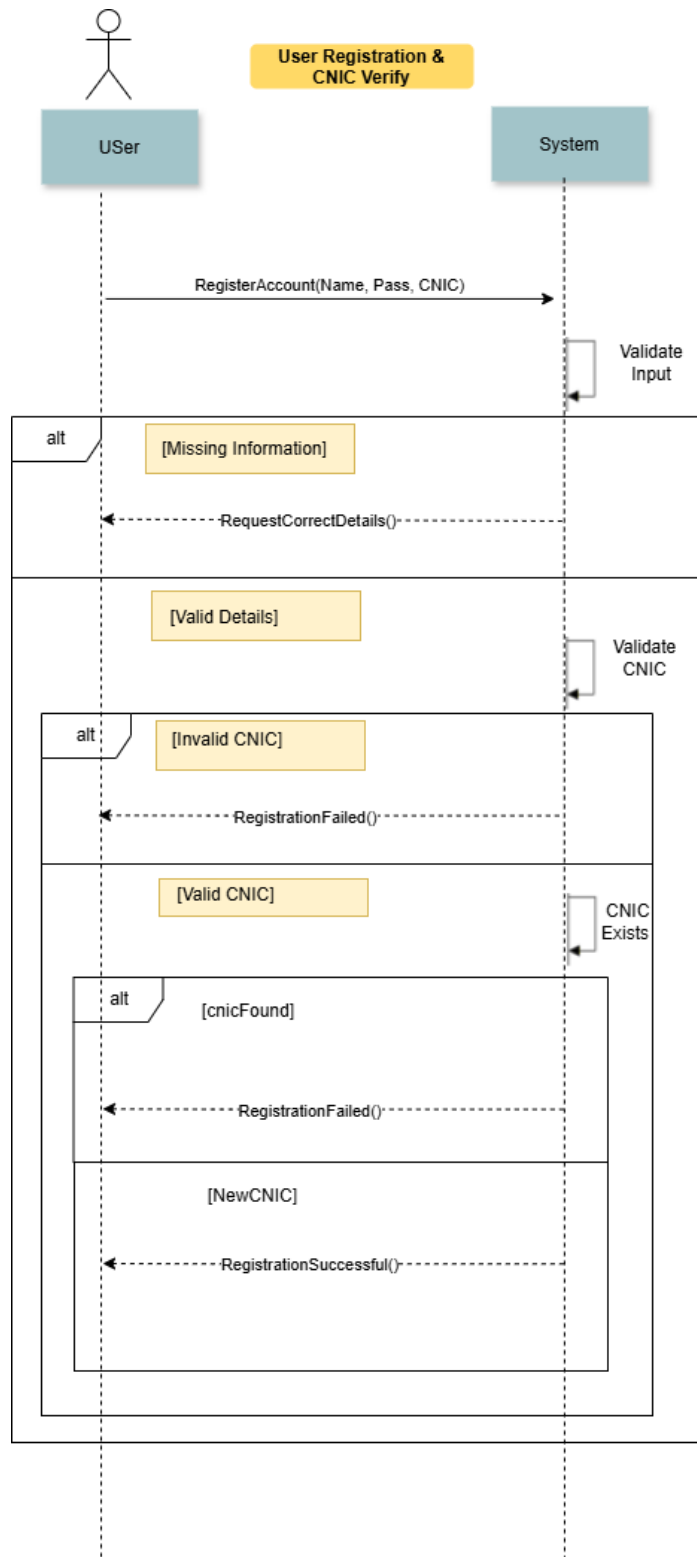


Figure 13: User Login/Register

9.1. Doctor Appointment

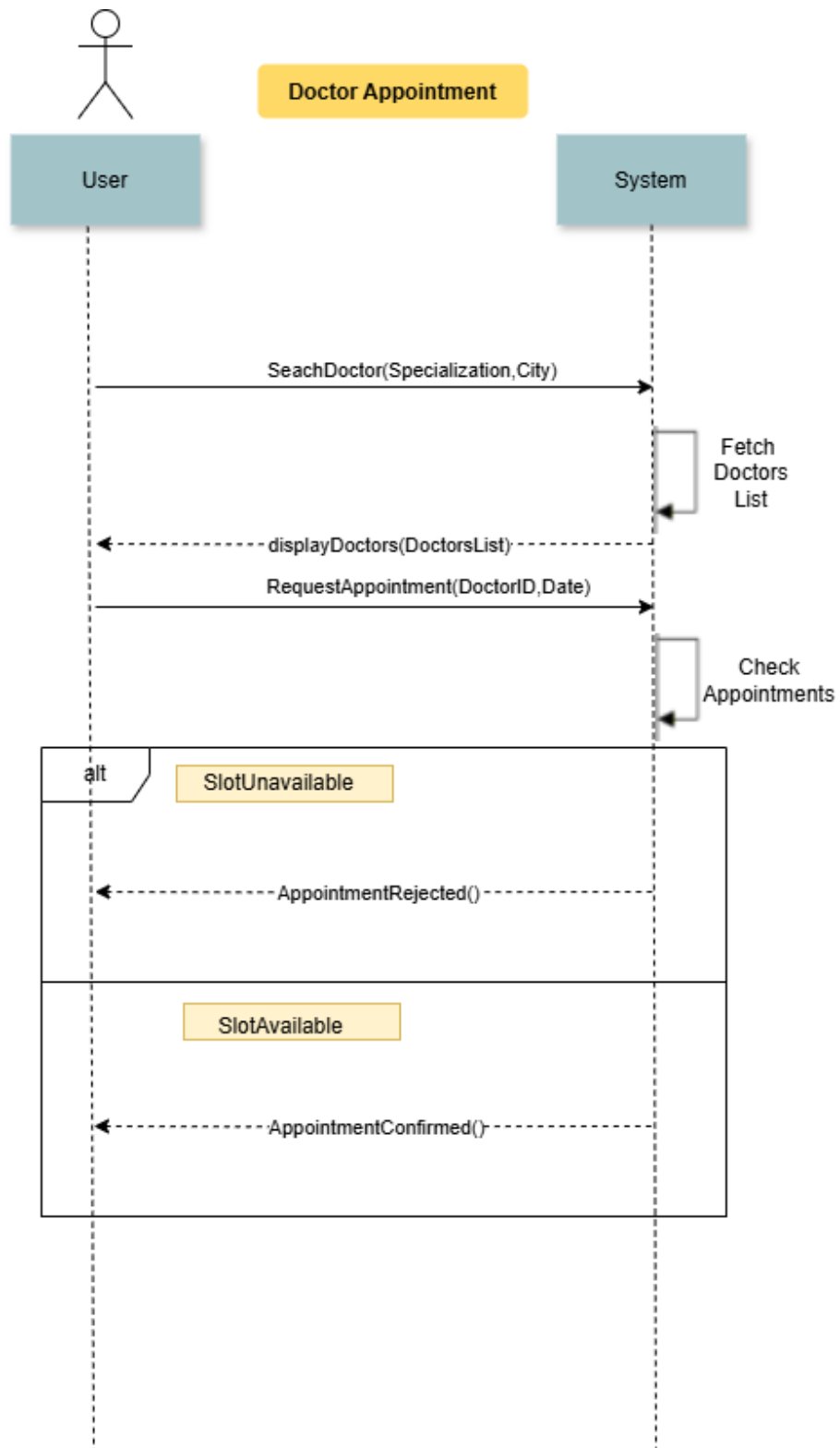


Figure 14: Doctor Appointment

9.2. Prescribe Test

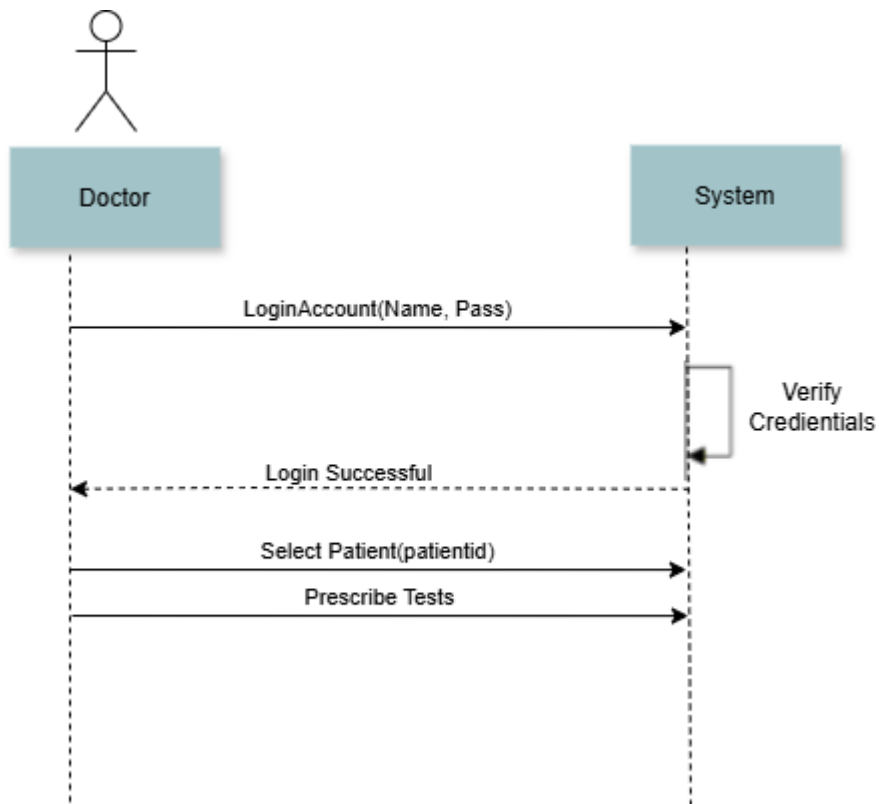


Figure 15: Prescribe Test

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9.3. Book Lab Test

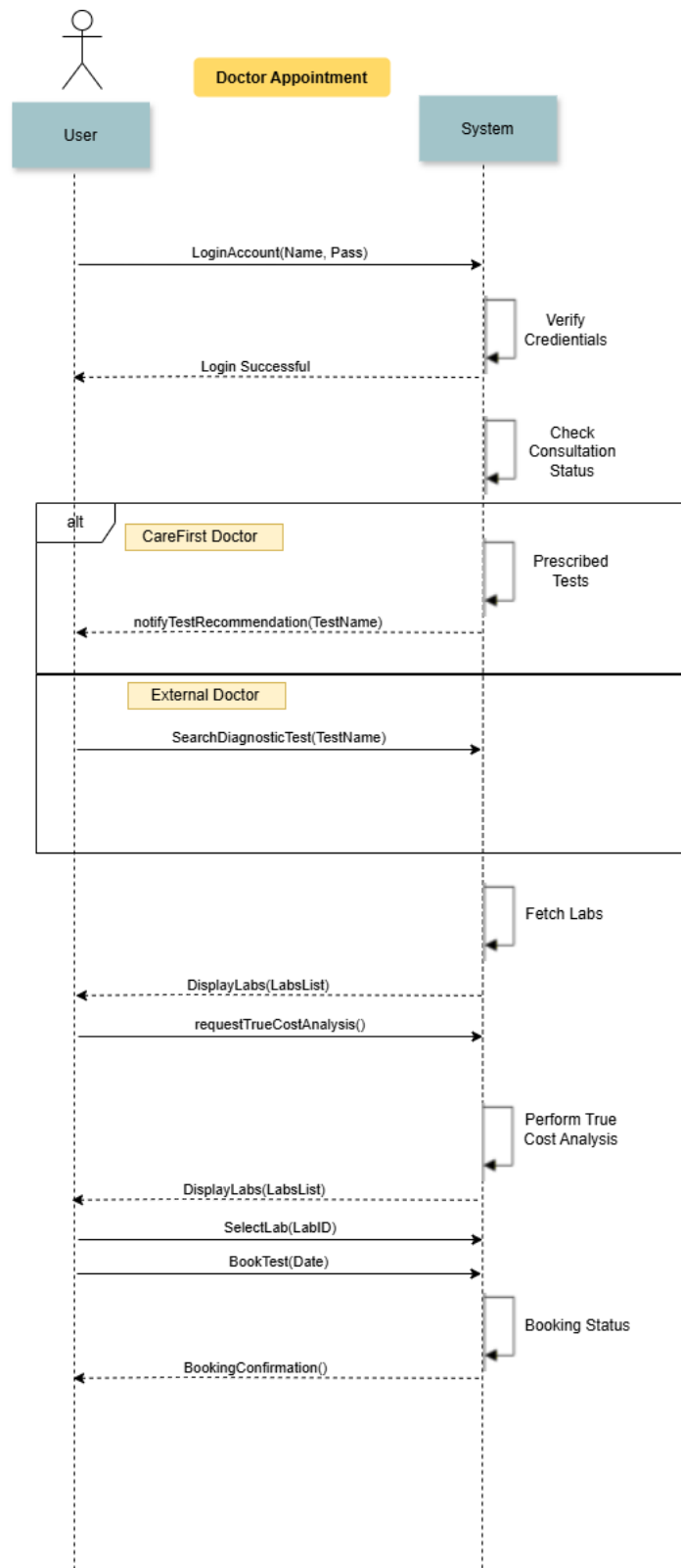


Figure 16: Book Lab Test

9.4. True Cost Analysis

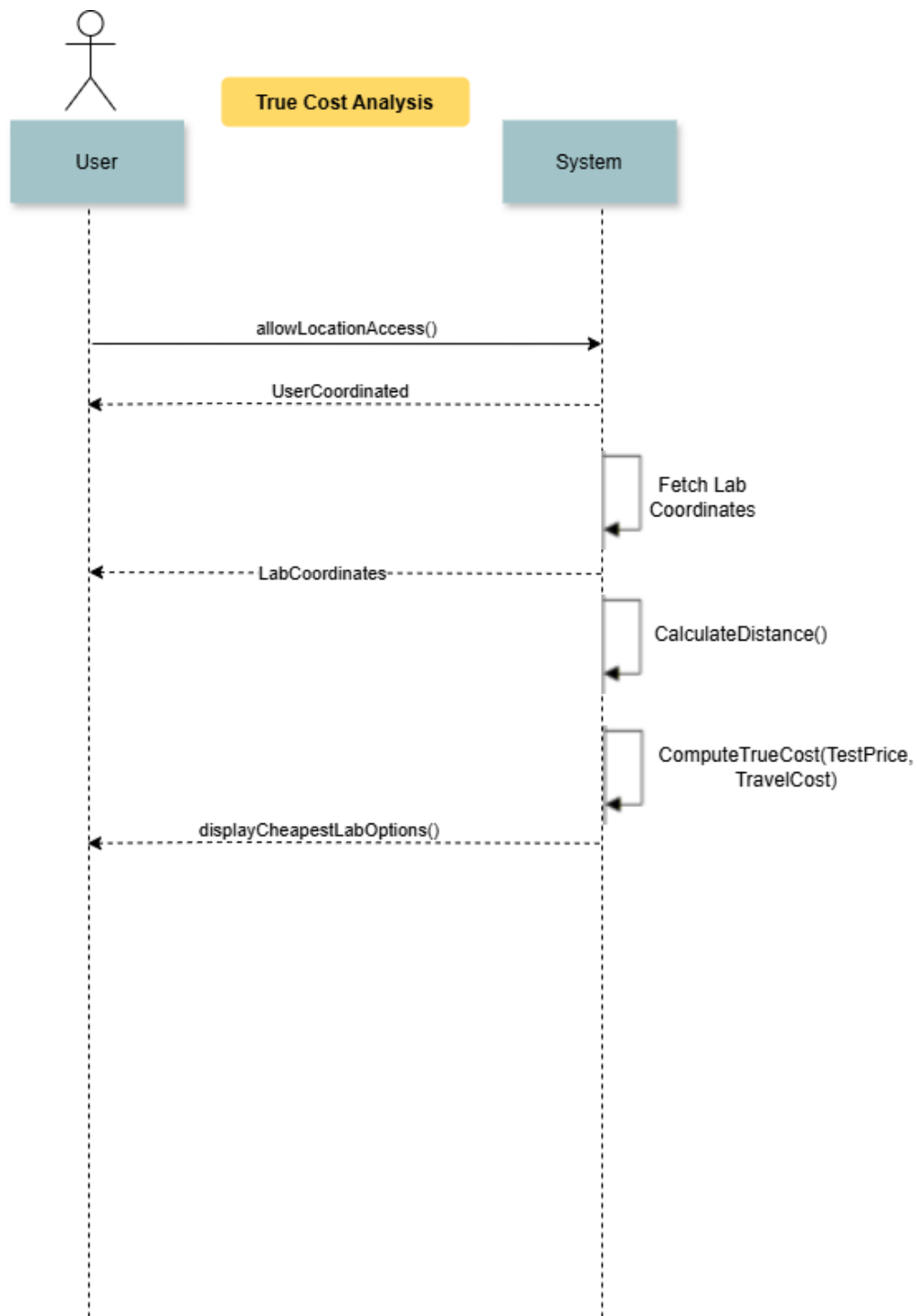


Figure 17: True Cost Analysis

9.5. Installment Plan

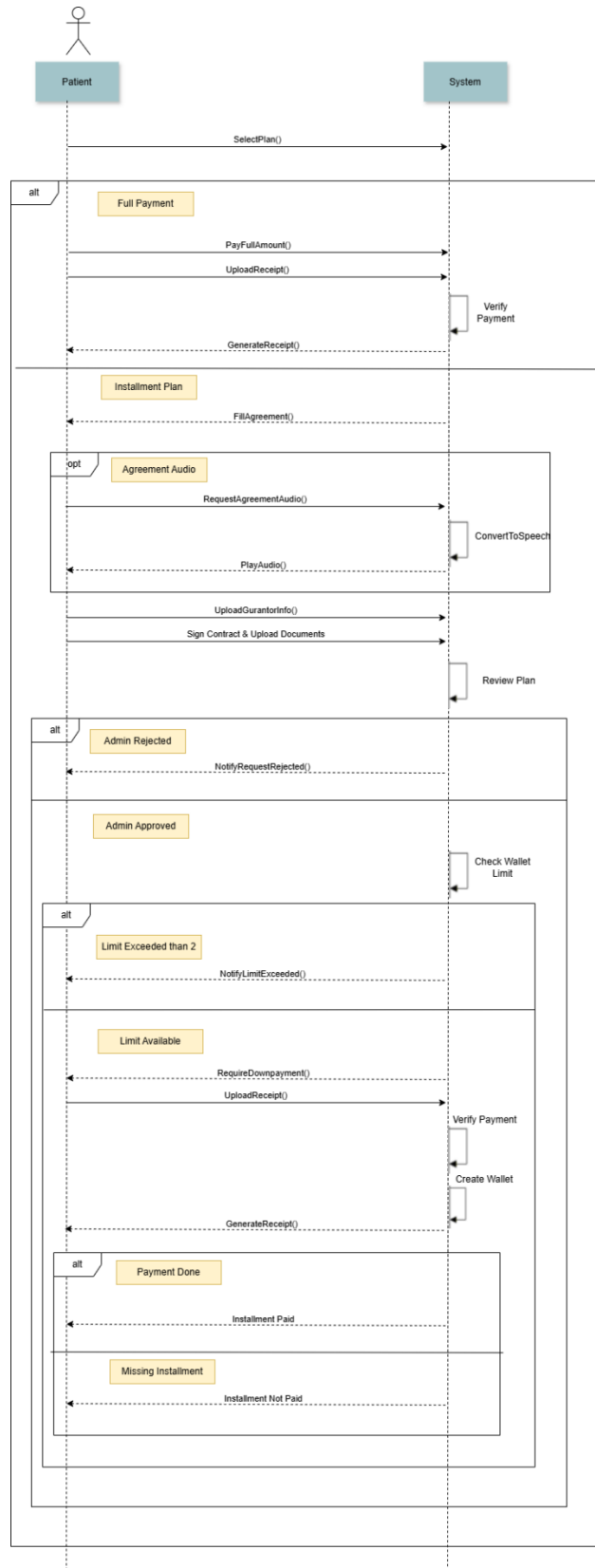


Figure 18: Installment Plan

9.6. Make Payment

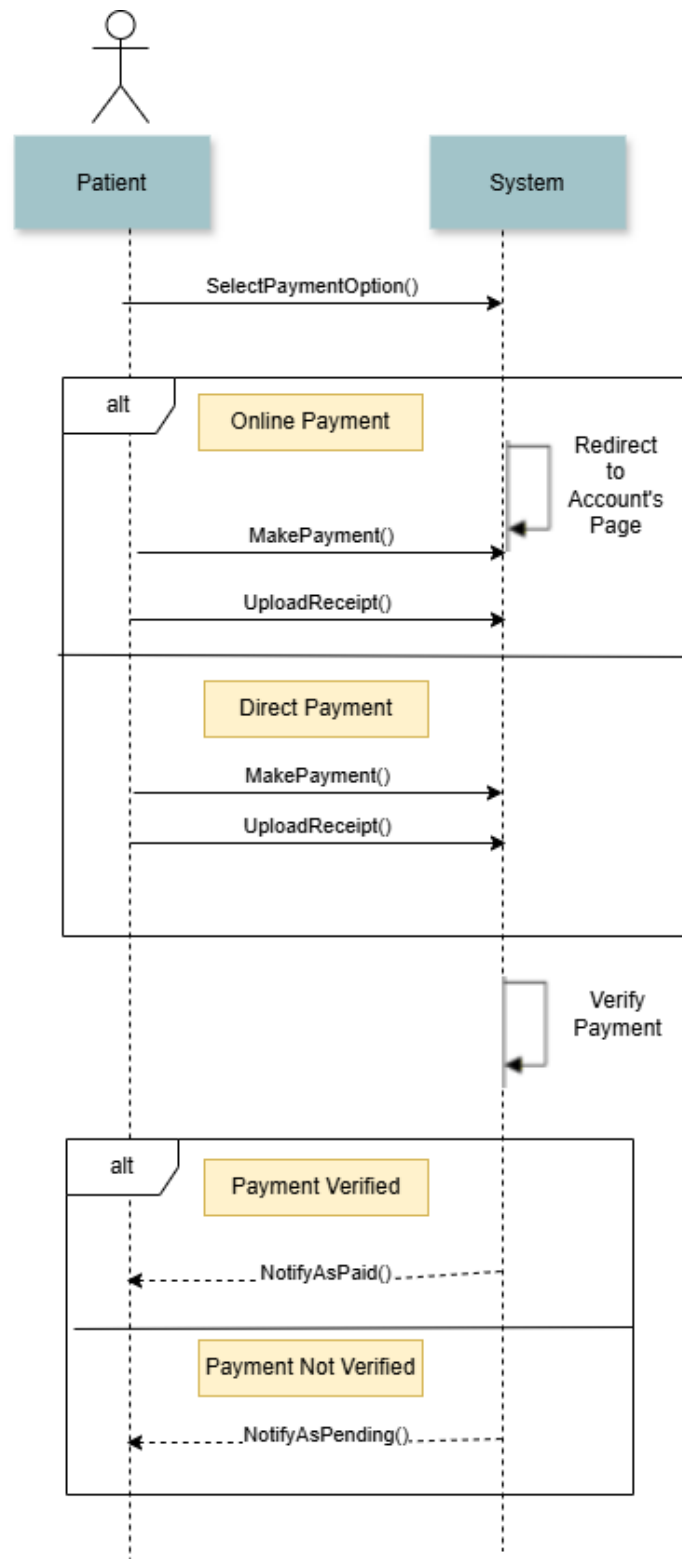
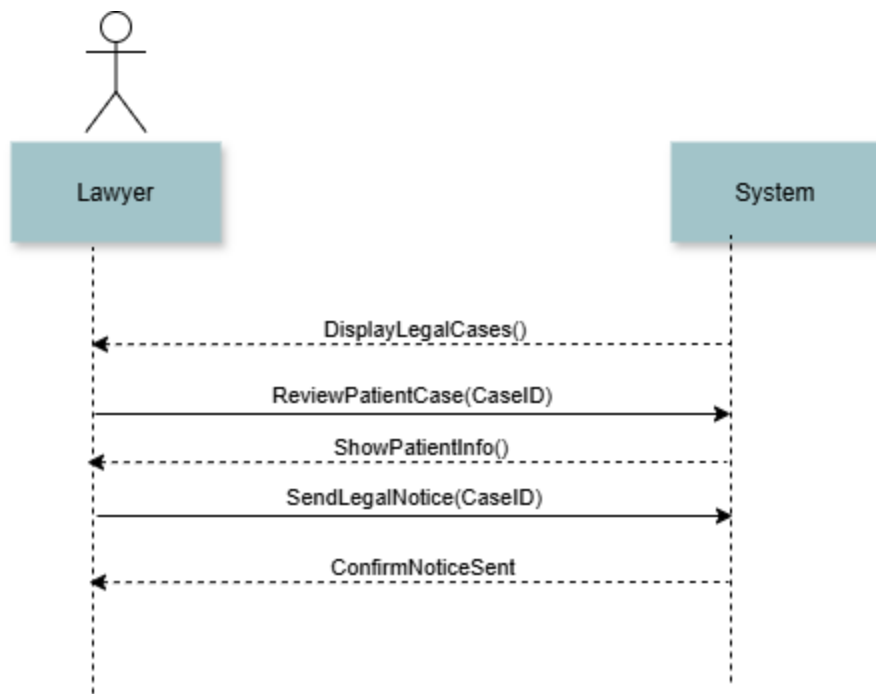


Figure 19: Make Payment

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9.7. Manage Defaulter Case

*Figure 20: Manage Defaulter Case*

9.8. View Report

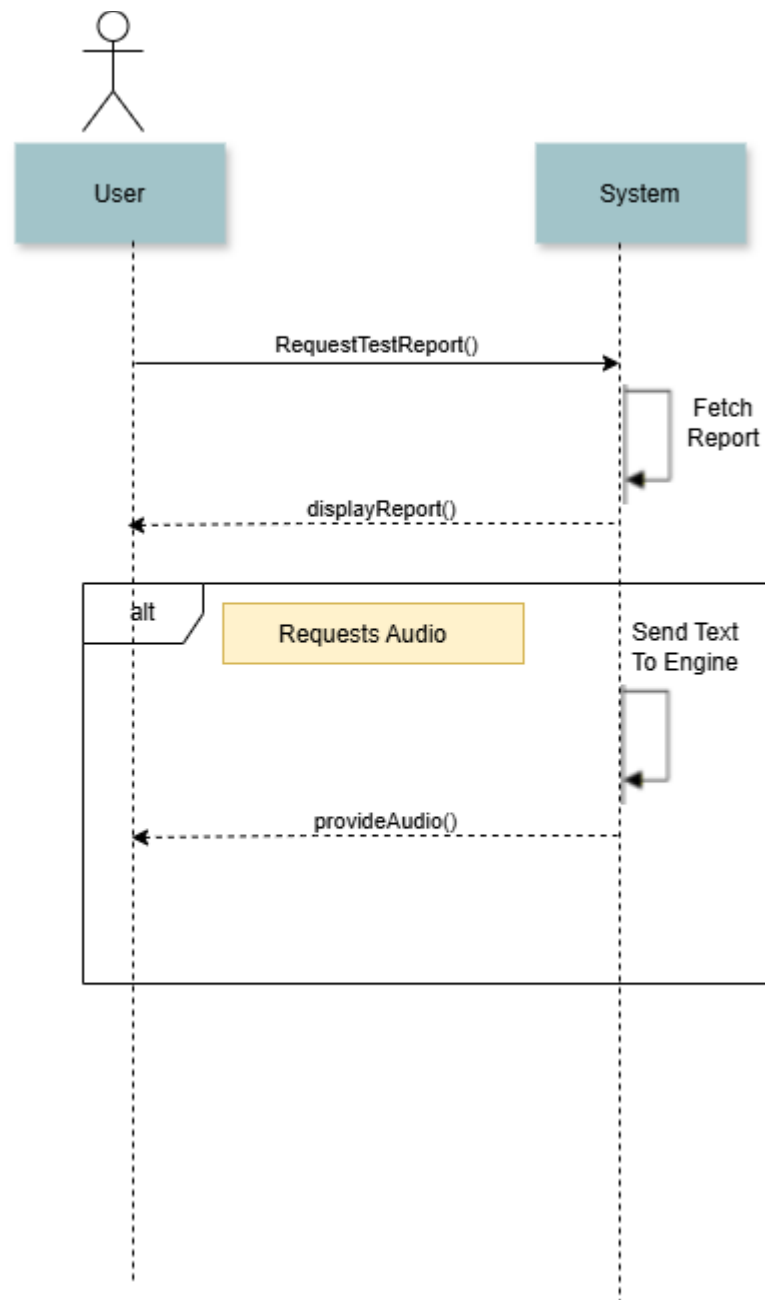


Figure 21: View Report

9.9. Full Funded Treatment

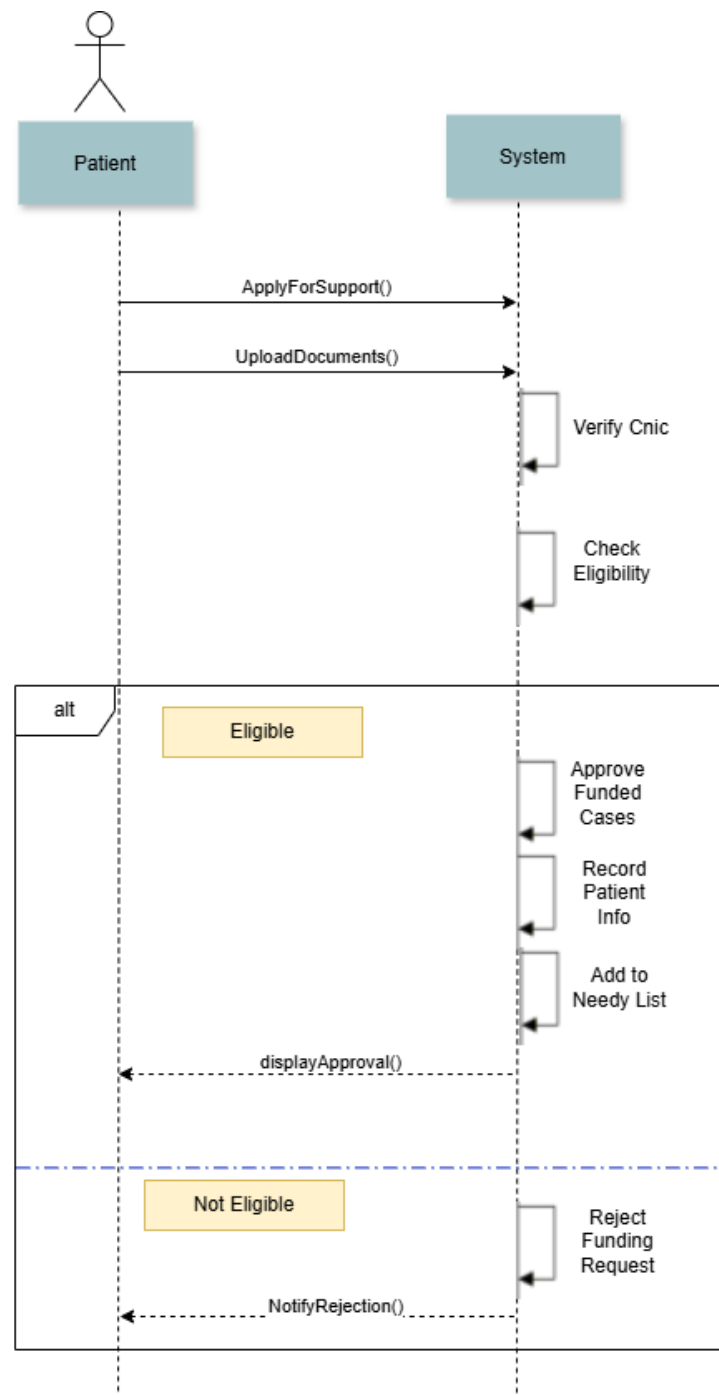
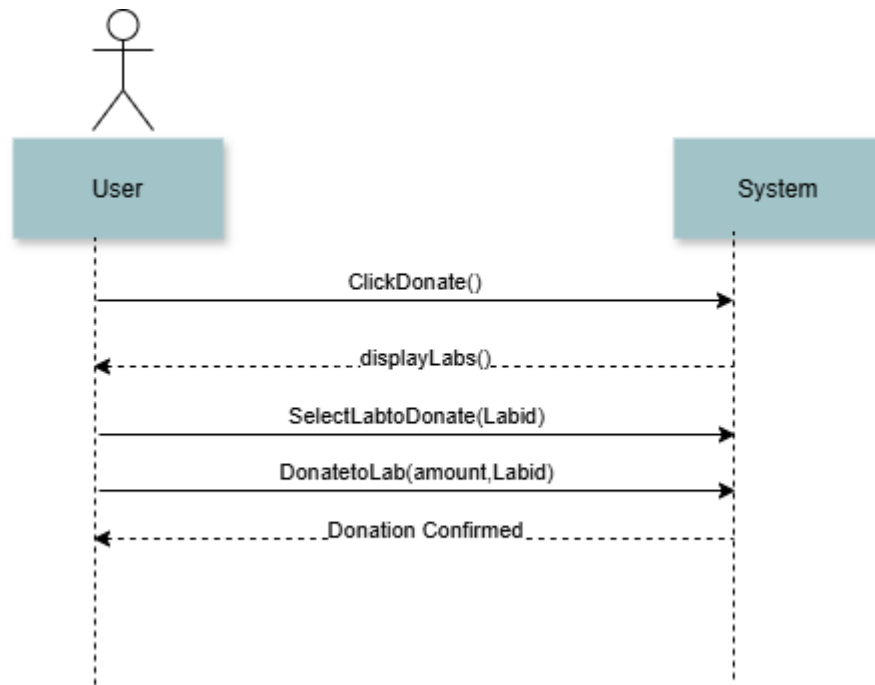


Figure 22: Full Funded Treatment

9.10. Donate

*Figure 23: Donate*

9.11. Manage Test

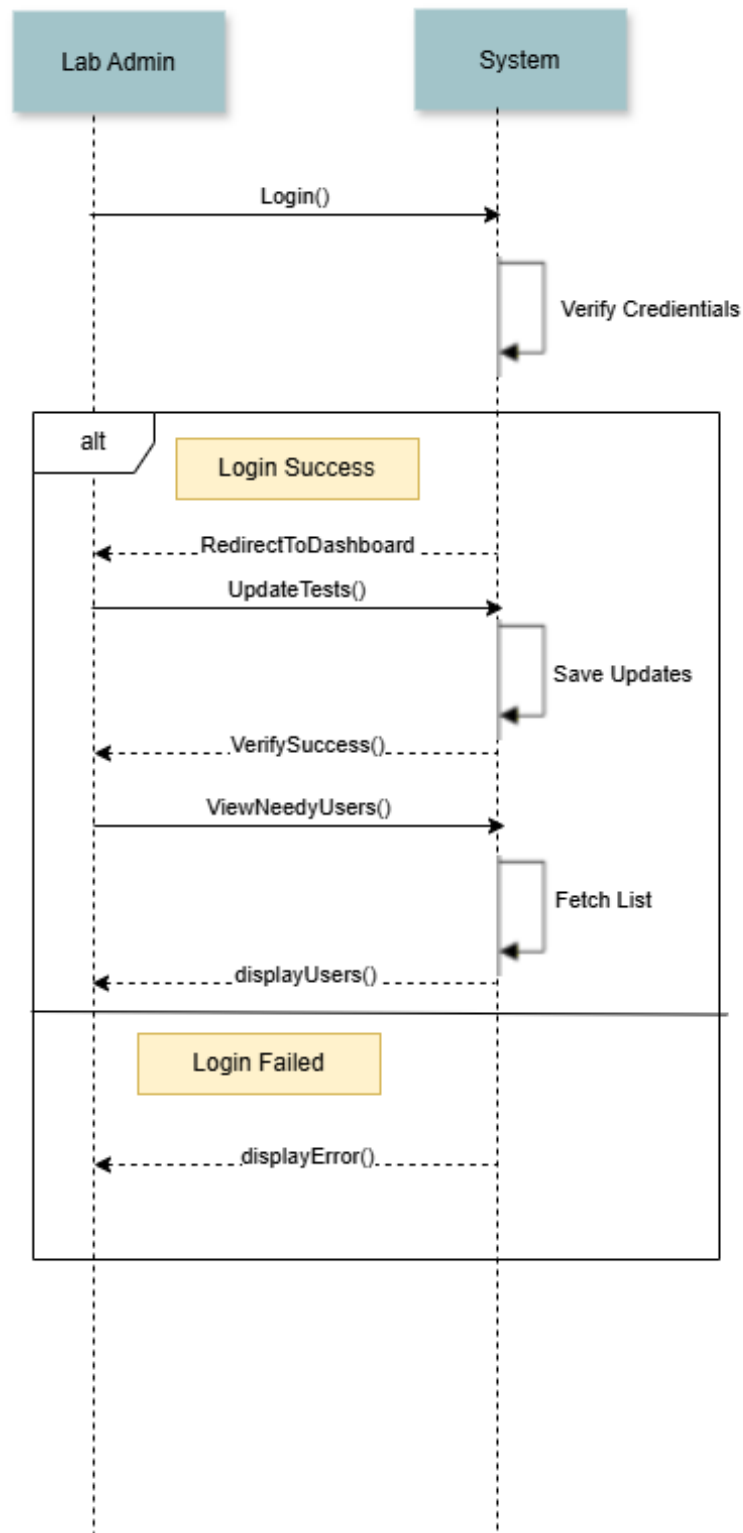


Figure 24: Manage Test

Figure 25: Class Diagram



12.0. Domain Model Diagram

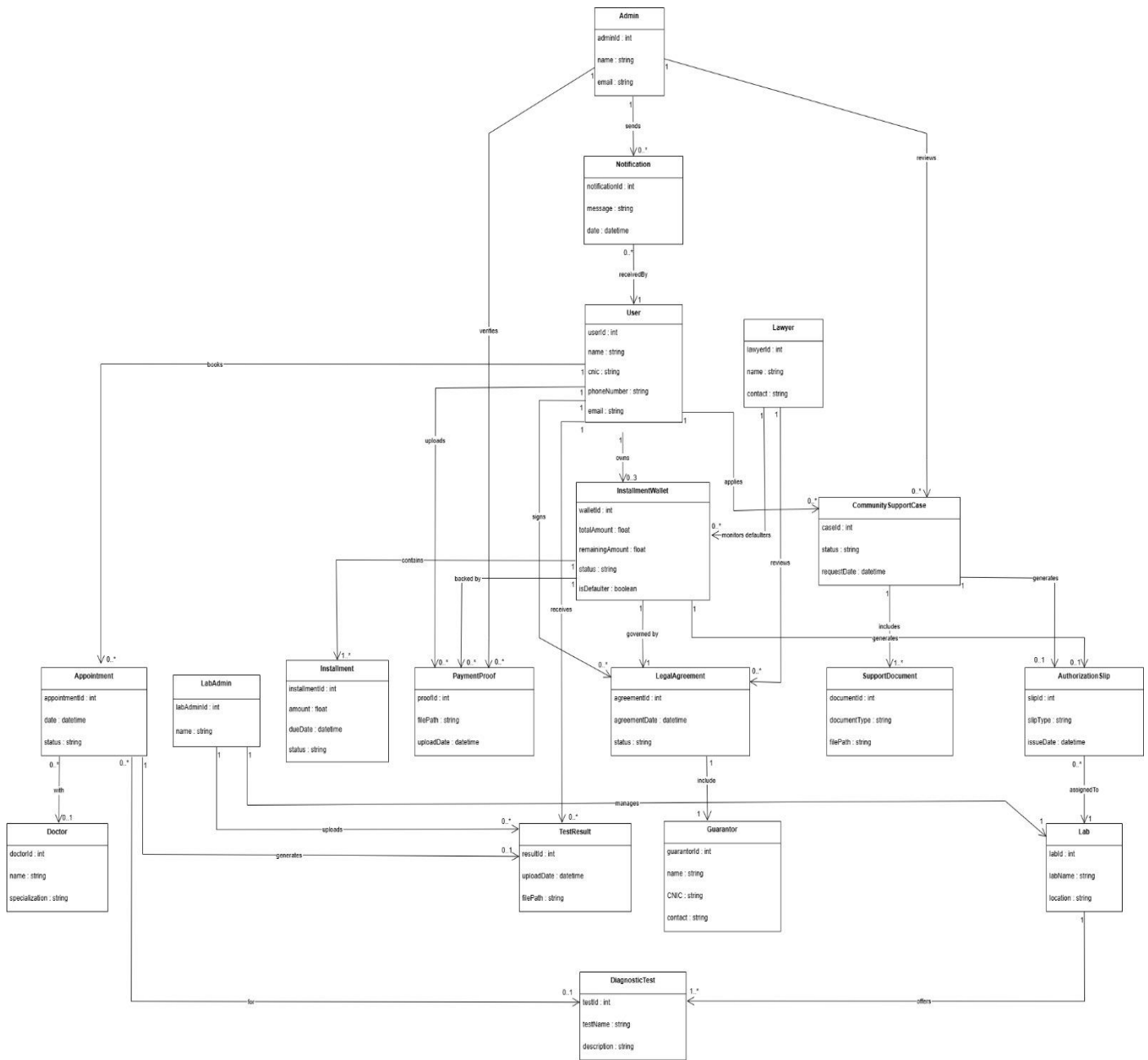


figure 27: Domain Model Diagram

13. Data Flow Diagram

13.0. DFD Level 0

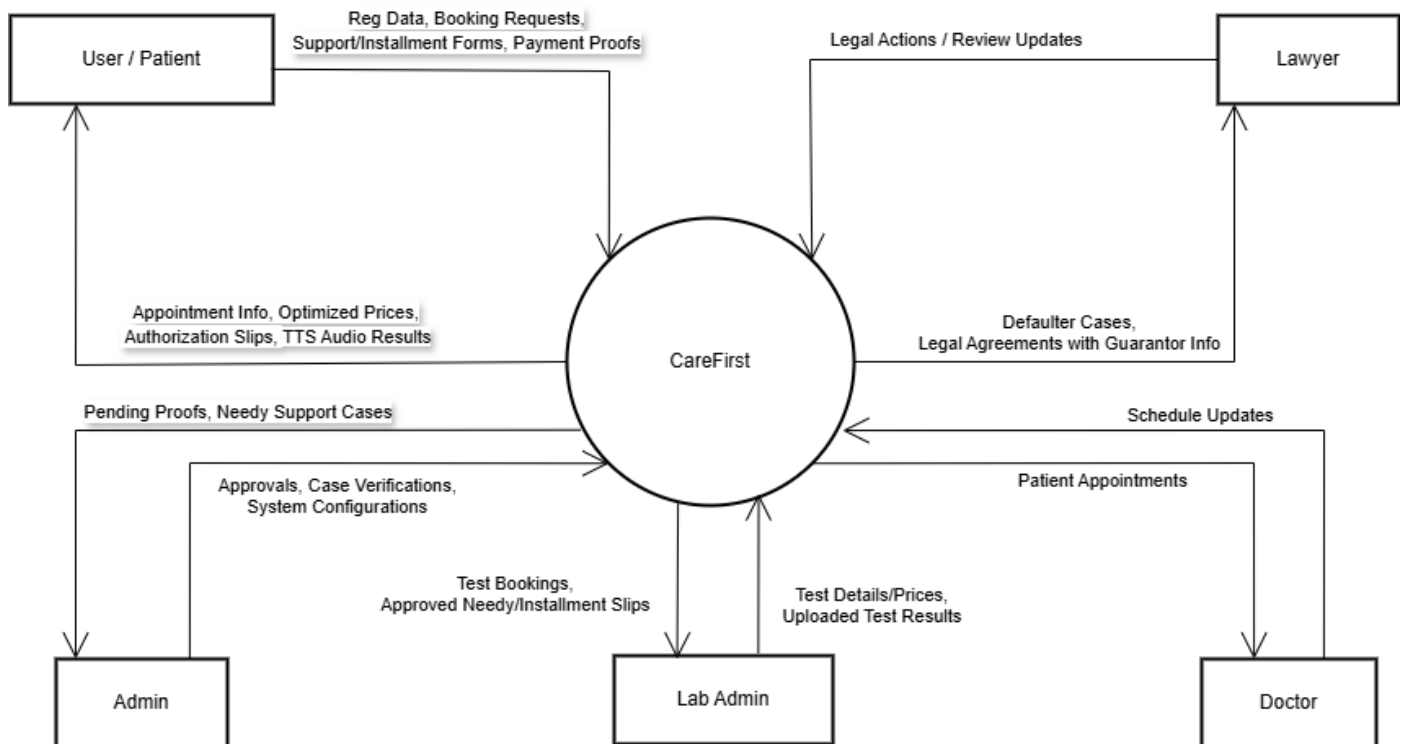


figure 28: DFD Level 0

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13.1. DFD Level 1

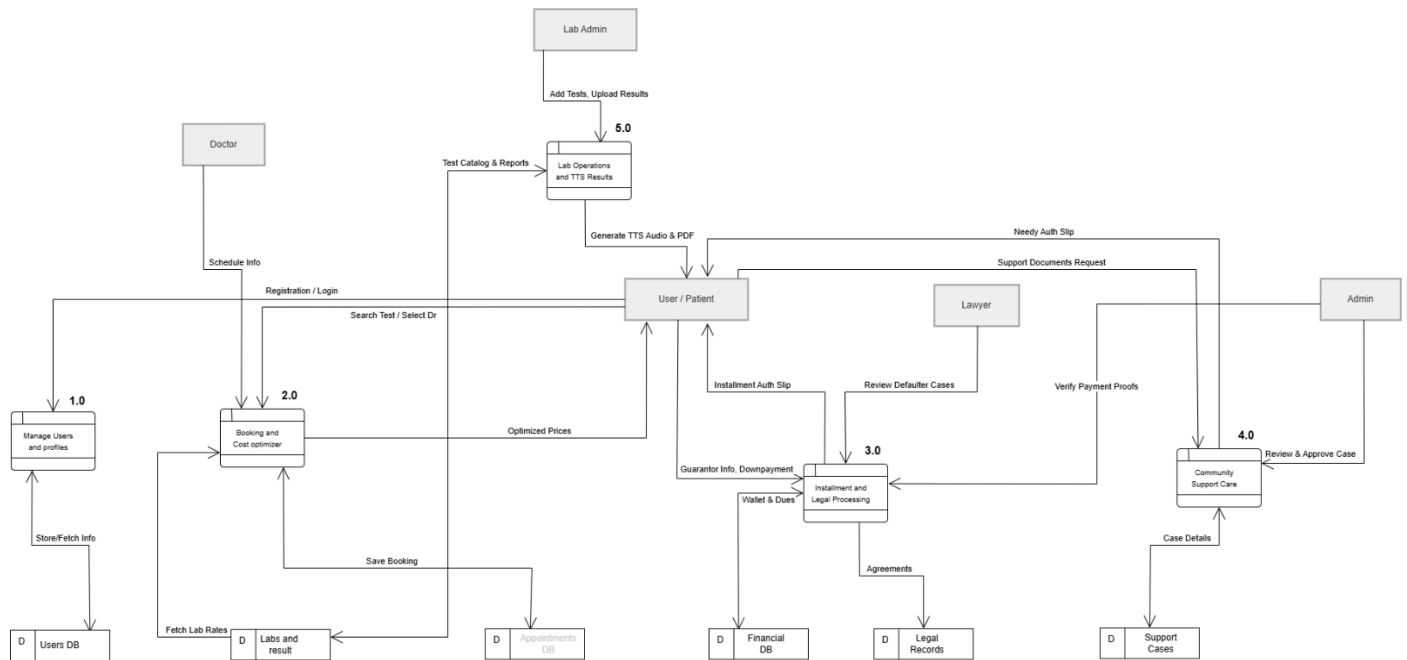


figure 29: DFD Level 1

13.2. DFD Level 2

13.2.1 Manage Users

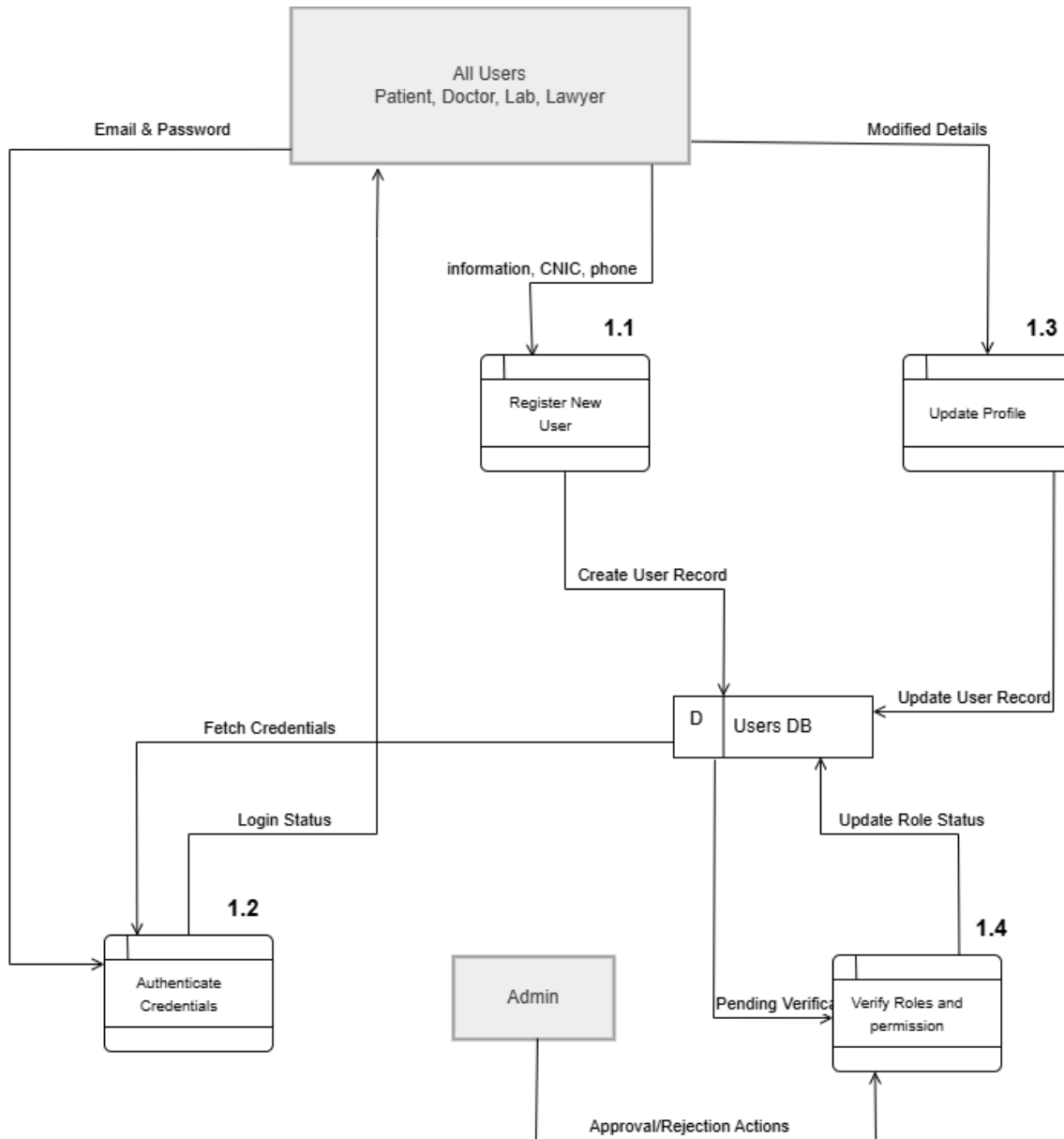


figure 30: Manage Users

13.2.2. Booking & Cost Optimizer

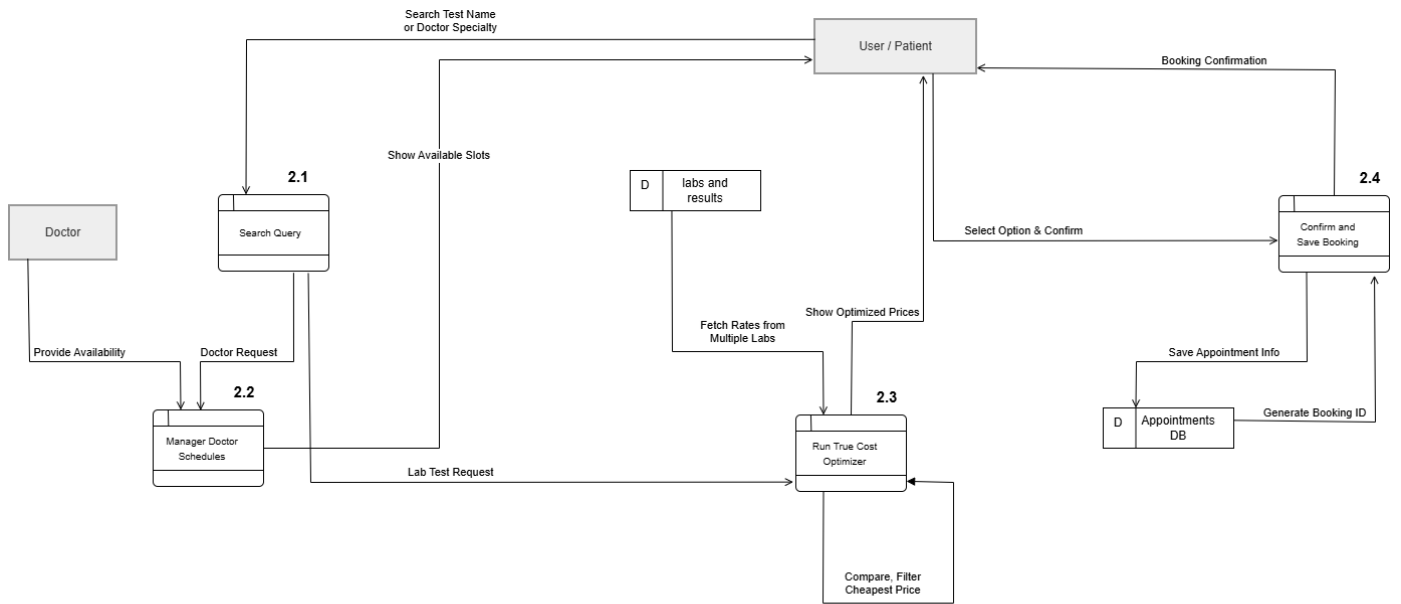


figure 31: Booking & Cost Optimizer

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13.2.3. Installment & Legal Handling

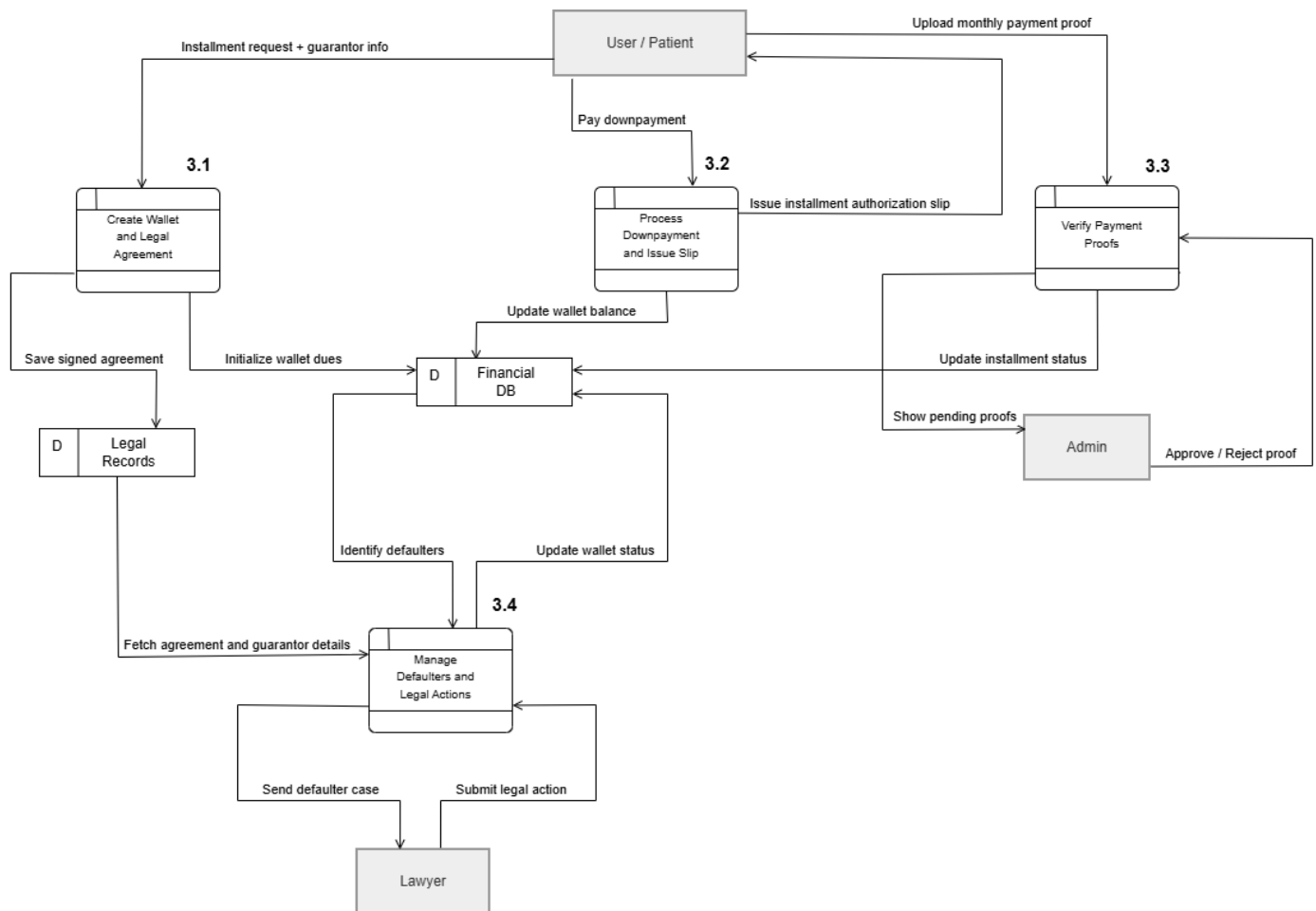


figure 32: Installment & Legal Handling

13.2.4. Support Case

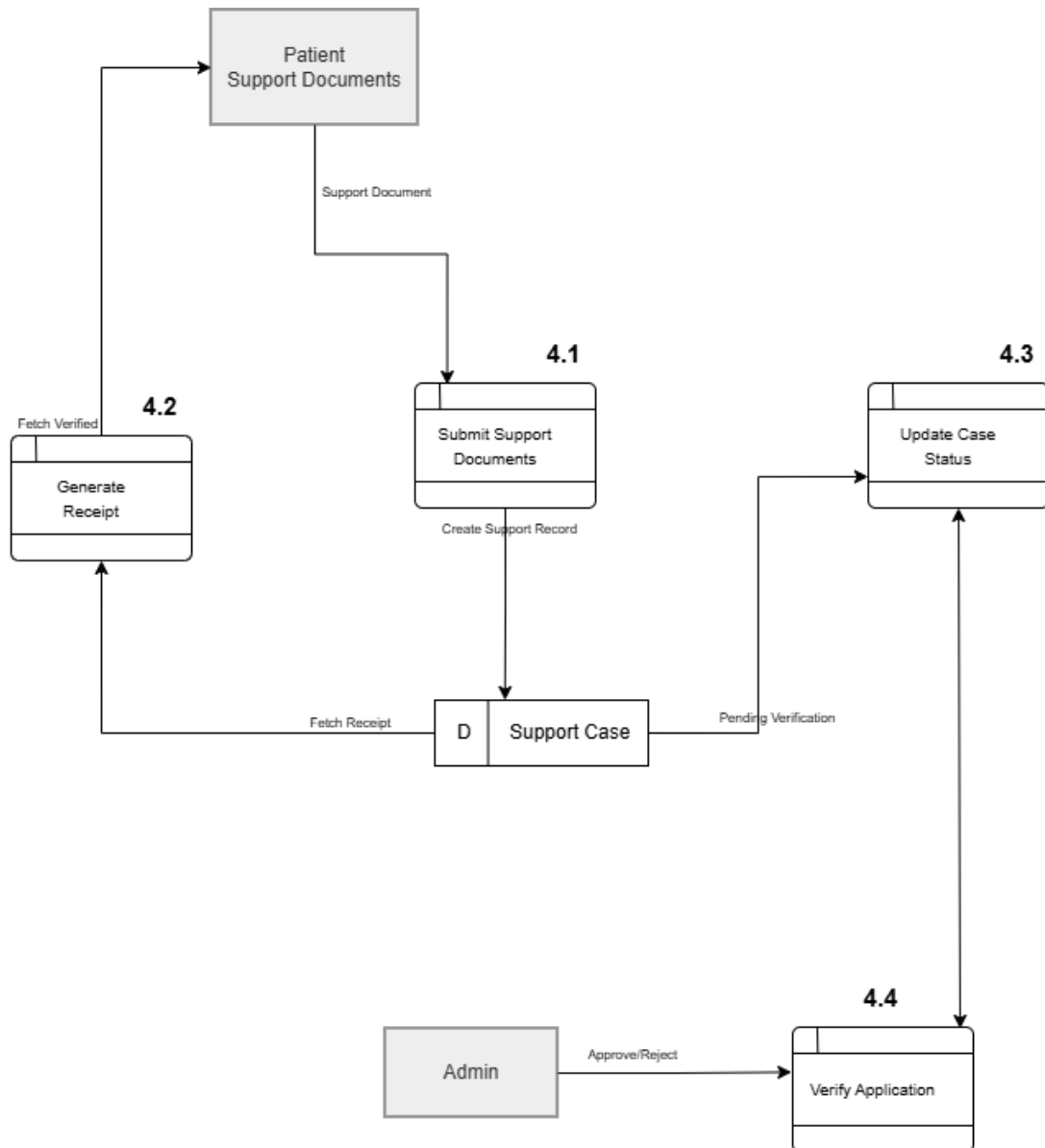


figure 32: Support Case

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13.2.5. TTS Engine

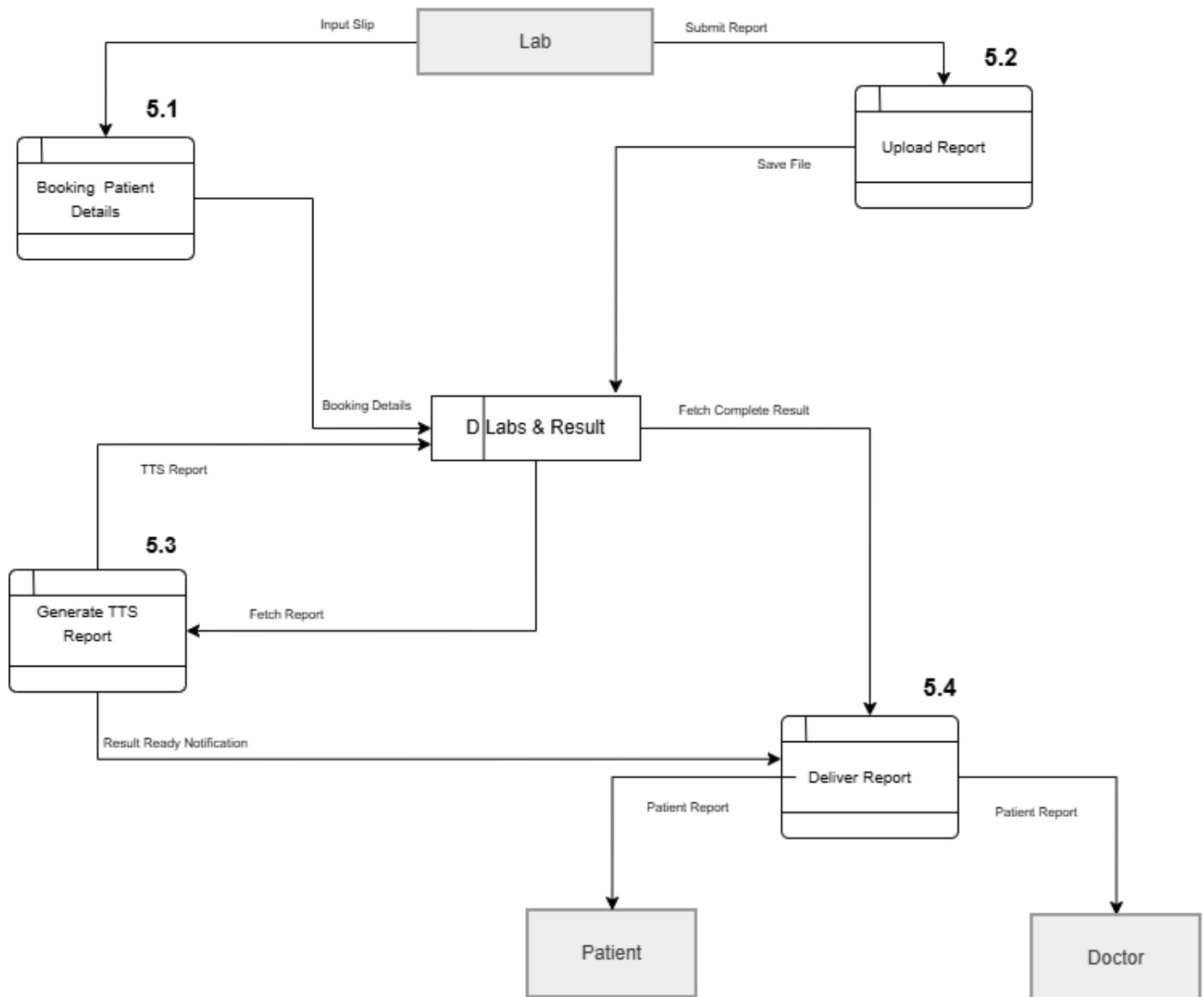


figure 33: TTS Engine

14. Sequence Diagram

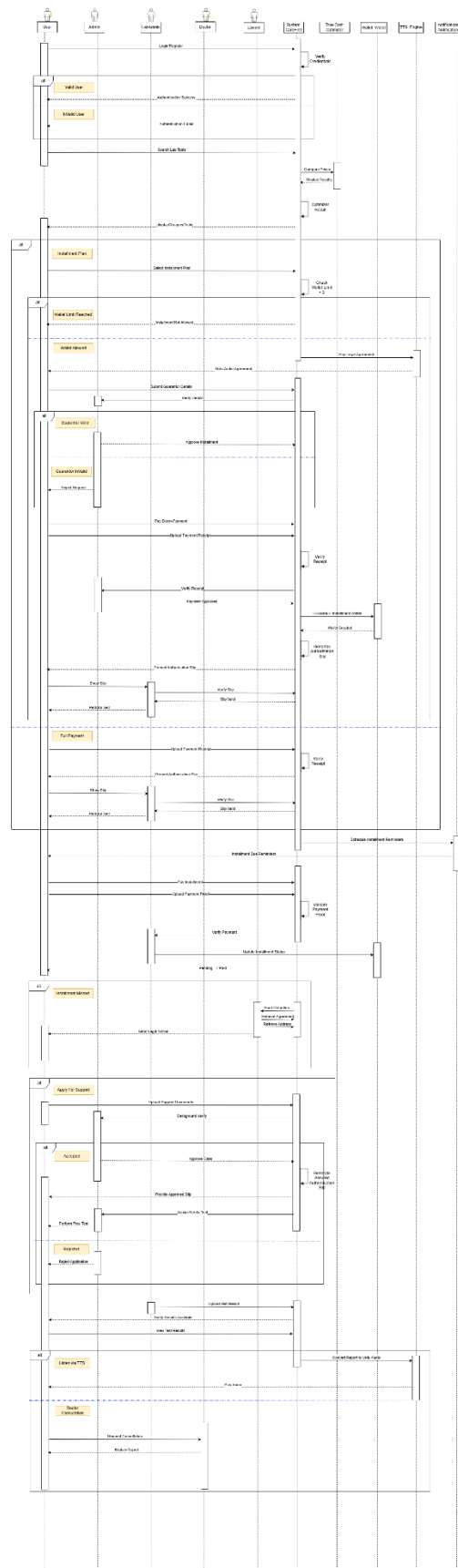


Figure 34: Sequence Diagram

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15. Architecture Diagram

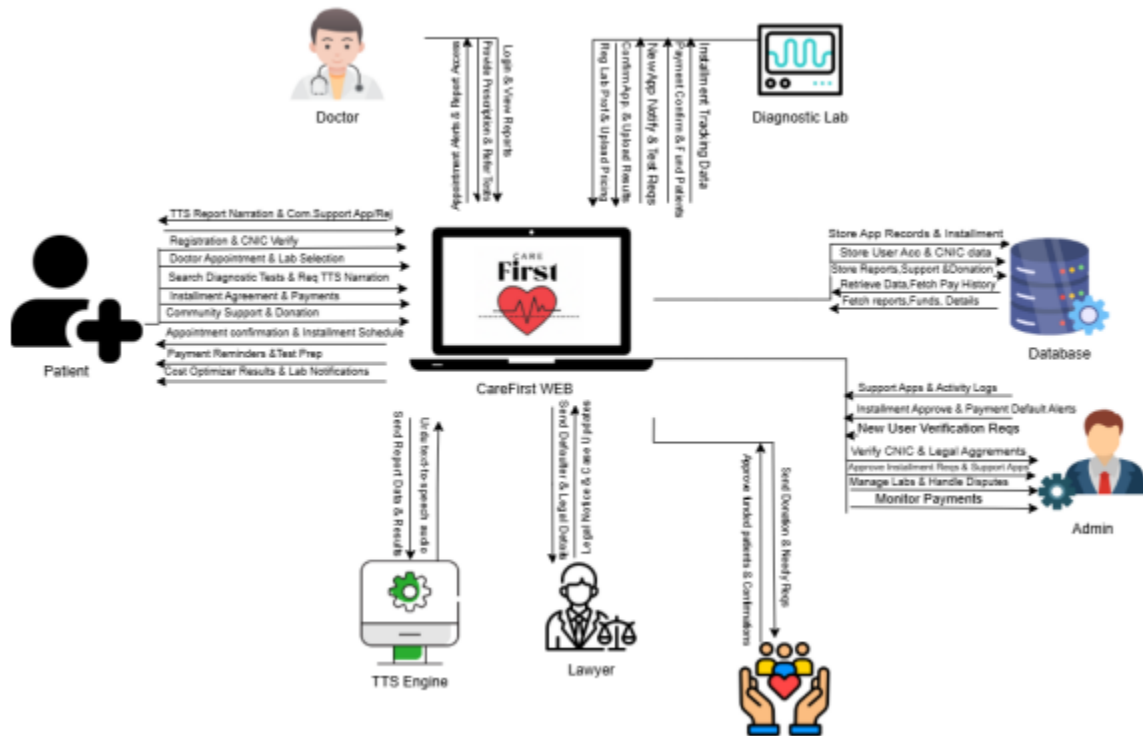


Figure 35: Architecture Diagram

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- [1] C. Larman, *Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development*, 3rd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2004, ch. 6, "Use Cases and Use Case Diagrams."
- [2] C. Larman, *Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development*, 3rd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2004, ch. 10, "System Sequence Diagrams."
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